

# CLASS 9 MATHEMATICS FOUNDATION

*Complete Exam-Ready Notes*

Structured with Quick Summaries · Real-Life Examples · Formula Sheets

Memory Hooks · Common Mistakes · Chapter-End Revision

# Chapter 1: Number Systems

## Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Familiarity with natural numbers, whole numbers and integers.
- Ability to convert a fraction into a decimal.
- Basic knowledge of squares, square roots and the number line.
- Understanding of laws of exponents for integer powers.

## 1.1 Rational and Irrational Numbers

 **Estimated time:** 15 mins   **Difficulty:**  Medium   ★★☆☆   **Exam Importance:** ★★★★★

### Quick Summary

A rational number can be written as  $p/q$  with  $q \neq 0$  and has a terminating or repeating decimal, while an irrational number cannot and has a non-terminating non-repeating decimal.

### Real-Life Example

The ratio of any circle's circumference to its diameter is  $\pi$ , a number no fraction can ever express exactly — which is why we always approximate it.

## Detailed Explanation

### Definition

A rational number is one that can be expressed in the form  $p/q$ , where  $p$  and  $q$  are integers and  $q \neq 0$ . An irrational number cannot be expressed in this form. Together they form the real numbers.

### Key Idea

The decimal expansion is the practical test. Terminating or recurring means rational; non-terminating and non-recurring means irrational. Every real number is one or the other, never both.

### Working Principle

A rational number  $p/q$  in lowest terms has a terminating decimal exactly when the denominator's prime factors are only 2s and 5s. Otherwise the decimal recurs.

### Applications

- $\sqrt{2}$ ,  $\sqrt{3}$ ,  $\sqrt{5}$ ,  $\pi$  and  $e$  are irrational;  $0.101001000100001\dots$  is irrational by construction.
- There are infinitely many rational and irrational numbers between any two distinct real numbers.
- $\sqrt{2}$  can be located on the number line using the Pythagoras theorem on a unit square.

### ✓ Points to Remember (Exam Focus)

- ✓ Rational: terminating or non-terminating recurring decimal.
- ✓ Irrational: non-terminating and non-recurring decimal.
- ✓  $p/q$  terminates only if the denominator in lowest terms is of the form  $2^m 5^n$ .
- ✓ Sum or product of a rational and an irrational number is always irrational.
- ✓ The sum or product of two irrational numbers may be rational, e.g.  $\sqrt{2} \times \sqrt{2} = 2$ .

### ⚠ Common Student Mistakes

- ⚠ Calling  $22/7$  irrational; it is rational and only an approximation to  $\pi$ .
- ⚠ Assuming the sum of two irrational numbers must be irrational —  $(2 + \sqrt{3}) + (2 - \sqrt{3}) = 4$ .
- ⚠ Writing  $\sqrt{4}$  as irrational; it equals 2 and is rational.

### 🧠 Memory Hook

'Repeats or Ends = Rational; Rambles forever = Irrational.'

### 🚀 Advanced Insights

Irrational numbers vastly outnumber rational ones: the rationals are countable while the reals are not. If you could pick a real number at random, the probability of getting a rational is exactly zero.

## Formula Section

### 📐 Rationalising the Denominator

|                         |                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------|
| Difficulty              | ● Medium ★★☆☆                                                                         |
| Formula                 | $1/(a + \sqrt{b}) \times (a - \sqrt{b})/(a - \sqrt{b}) = (a - \sqrt{b})/(a^2 - b)$    |
| Meaning of Variables    | $a$ = rational part, $\sqrt{b}$ = irrational part of the denominator.                 |
| When to Use             | Whenever a surd appears in a denominator and must be removed.                         |
| Common Mistakes         | Changing the sign of the wrong term, or forgetting to multiply the numerator as well. |
| Shortcut / Memory Trick | Multiply by the conjugate — change the sign between the two terms.                    |

## 1.2 Laws of Exponents and Surds

🕒 Estimated time: 14 mins    Difficulty: ● Medium ★★☆☆    Exam Importance: ★★★★★

### Quick Summary

The laws of exponents extend from integer to rational powers, allowing roots to be written as fractional indices and simplified using the same rules.

### Real-Life Example

Compound interest, population growth and sound intensity are all computed with exponent laws, which is why they are worth complete fluency.

## Detailed Explanation

### Definition

For a positive real number  $a$  and rational numbers  $m$  and  $n$ ,  $a^m \times a^n = a^{m+n}$ . A surd is an irrational root of a rational number, such as  $\sqrt{2}$  or  $\sqrt[3]{5}$ .

### Key Idea

Writing a root as a fractional index —  $\sqrt[n]{a} = a^{1/n}$  — turns every root manipulation into an exponent manipulation, so one set of rules covers both.

### Working Principle

Surds are simplified by extracting perfect squares (or cubes) from under the radical, and like surds are added or subtracted just as like terms are.

### Applications

- $\sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2}$ .
- Rationalising  $1/\sqrt{3}$  gives  $\sqrt{3}/3$ .
- $a^{(m/n)} = \sqrt[n]{a^m}$  connects indices and roots directly.

### ✓ Points to Remember (Exam Focus)

- ✓  $a^m \times a^n = a^{m+n}$ ;  $a^m \div a^n = a^{m-n}$ ;  $(a^m)^n = a^{mn}$ ;  $a^0 = 1$  for  $a \neq 0$ .
- ✓  $a^{1/n} = \sqrt[n]{a}$  and  $a^{(m/n)} = \sqrt[n]{a^m}$ .
- ✓  $a^{-n} = 1/a^n$ .
- ✓ Only like surds can be added or subtracted.
- ✓ The laws hold for  $a > 0$ ; for negative bases with fractional powers they can fail.

### Common Student Mistakes

-  Writing  $(a + b)^n = a^n + b^n$ , which is false.
-  Adding unlike surds, e.g. treating  $\sqrt{2} + \sqrt{3}$  as  $\sqrt{5}$ .
-  Forgetting that  $a^0 = 1$  only when  $a \neq 0$ .

 **Memory Hook**

'Multiply → add powers; Divide → subtract powers; Power of a power → multiply.'

 **Advanced Insights**

Defining  $a^x$  for irrational  $x$  requires limits, which is where exponents connect to calculus. This is why the natural exponential  $e^x$ , defined by its own derivative, becomes the base of choice in higher mathematics.

**Formula Section** **Laws of Exponents**

|                                |                                                                                           |
|--------------------------------|-------------------------------------------------------------------------------------------|
| <b>Difficulty</b>              | ● Easy ★☆☆                                                                                |
| <b>Formula</b>                 | $a^m \cdot a^n = a^{m+n}$ , $a^m / a^n = a^{m-n}$ , $(a^m)^n = a^{mn}$ , $a^{-n} = 1/a^n$ |
| <b>Meaning of Variables</b>    | $a$ = base ( $a > 0$ ), $m$ and $n$ = rational exponents.                                 |
| <b>When to Use</b>             | For all simplification of powers, roots and surds.                                        |
| <b>Common Mistakes</b>         | Adding the exponents when raising a power to a power instead of multiplying them.         |
| <b>Shortcut / Memory Trick</b> | Same base multiplied → add; same base divided → subtract.                                 |

 **Fractional Exponent**

|                                |                                                                          |
|--------------------------------|--------------------------------------------------------------------------|
| <b>Difficulty</b>              | ● Medium ★★☆☆                                                            |
| <b>Formula</b>                 | $a^{(m/n)} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$                            |
| <b>Meaning of Variables</b>    | $a$ = positive base, $m$ = power, $n$ = root index.                      |
| <b>When to Use</b>             | To convert between root notation and index notation.                     |
| <b>Common Mistakes</b>         | Interchanging the roles of $m$ and $n$ .                                 |
| <b>Shortcut / Memory Trick</b> | The denominator of the fraction is the root; the numerator is the power. |

**II Quick Recap — Topics Covered So Far**

- Rational and Irrational Numbers — A rational number can be written as  $p/q$  with  $q \neq 0$  and has a terminating or repeating decimal, while an irrational number cannot and has a non-terminating non-repeating decimal.
- Laws of Exponents and Surds — The laws of exponents extend from integer to rational powers, allowing roots to be written as fractional indices and simplified using the same rules.



## Chapter 1 — End of Chapter Revision

### All Memory Hooks

- Rational and Irrational Numbers: 'Repeats or Ends = Rational; Rambles forever = Irrational.'
- Laws of Exponents and Surds: 'Multiply → add powers; Divide → subtract powers; Power of a power → multiply.'

### All Common Mistakes

-  Calling  $22/7$  irrational; it is rational and only an approximation to  $\pi$ .
-  Assuming the sum of two irrational numbers must be irrational —  $(2 + \sqrt{3}) + (2 - \sqrt{3}) = 4$ .
-  Writing  $\sqrt{4}$  as irrational; it equals 2 and is rational.
-  Writing  $(a + b)^n = a^n + b^n$ , which is false.
-  Adding unlike surds, e.g. treating  $\sqrt{2} + \sqrt{3}$  as  $\sqrt{5}$ .
-  Forgetting that  $a^0 = 1$  only when  $a \neq 0$ .

### Formula Sheet

| Formula                                                                                                                  | Meaning / Use                                                 |
|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Rationalising the Denominator:</b> $1/(a + \sqrt{b}) \times (a - \sqrt{b})/(a - \sqrt{b}) = (a - \sqrt{b})/(a^2 - b)$ | Whenever a surd appears in a denominator and must be removed. |
| <b>Laws of Exponents:</b> $a^m \cdot a^n = a^{m+n}$ , $a^m/a^n = a^{m-n}$ , $(a^m)^n = a^{mn}$ , $a^{-n} = 1/a^n$        | For all simplification of powers, roots and surds.            |
| <b>Fractional Exponent:</b> $a^{(m/n)} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$                                                | To convert between root notation and index notation.          |

### Quick Revision Section

- Rational and Irrational Numbers: A rational number can be written as  $p/q$  with  $q \neq 0$  and has a terminating or repeating decimal, while an irrational number cannot and has a non-terminating non-repeating decimal.
- Laws of Exponents and Surds: The laws of exponents extend from integer to rational powers, allowing roots to be written as fractional indices and simplified using the same rules.

## Chapter 2: Polynomials

### Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Comfort with algebraic expressions and like terms.
- Knowledge of the laws of exponents.
- Ability to substitute a value into an expression.
- Familiarity with factorisation of simple expressions.

### 2.1 Polynomials, Degree and Zeroes

 **Estimated time:** 14 mins   **Difficulty:**  Easy   ★☆☆   **Exam Importance:** ★★★★★

#### Quick Summary

A polynomial is an algebraic expression in which the powers of the variable are whole numbers, and a zero of a polynomial is a value of the variable that makes it equal to zero.

#### Real-Life Example

The path of a thrown ball is described by a quadratic polynomial, and its zeroes are exactly the two moments it is at ground level.

#### Detailed Explanation

##### Definition

A polynomial in one variable  $x$  is an expression of the form  $a_n x^n + \dots + a_1 x + a_0$  where the coefficients are real numbers and  $n$  is a whole number. The degree is the highest power of the variable appearing with a non-zero coefficient.

##### Key Idea

The powers must be whole numbers. Expressions such as  $x^{1/2}$  or  $1/x$  are not polynomials, however algebraic they look — this is the single most common classification error.

##### Working Principle

A number  $k$  is a zero of  $p(x)$  if  $p(k) = 0$ . A polynomial of degree  $n$  has at most  $n$  zeroes, and a linear polynomial has exactly one.

##### Applications

- Linear (degree 1), quadratic (degree 2), cubic (degree 3).
- The zero polynomial has no degree defined; a non-zero constant has degree 0.
- Zeroes of a polynomial are the  $x$ -coordinates where its graph meets the  $x$ -axis.

**✓ Points to Remember (Exam Focus)**

- ✓ Powers of the variable must be whole numbers for an expression to be a polynomial.
- ✓ Degree = highest power with a non-zero coefficient.
- ✓ A polynomial of degree  $n$  has at most  $n$  zeroes.
- ✓ A linear polynomial  $ax + b$  has exactly one zero,  $x = -b/a$ .
- ✓ The degree of the zero polynomial is not defined.

**⚠ Common Student Mistakes**

- ⚠ Calling  $\sqrt{x} + 3$  or  $1/x + 2$  a polynomial; fractional and negative powers are not allowed.
- ⚠ Saying the degree of a non-zero constant is undefined; it is 0.
- ⚠ Confusing the value of a polynomial  $p(k)$  with a zero of the polynomial.

**Memory Hook**

**'Whole-number powers only' — that one rule decides whether an expression is a polynomial.**

**Advanced Insights**

The Fundamental Theorem of Algebra guarantees that a degree- $n$  polynomial has exactly  $n$  roots when complex numbers are allowed. The 'at most  $n$ ' rule of school algebra becomes 'exactly  $n$ ' once the number system is complete.

## 2.2 Remainder Theorem and Factor Theorem

🕒 **Estimated time:** 15 mins   **Difficulty:** ● Medium   ★★☆☆   **Exam Importance:** ★★★★★

**Quick Summary**

The remainder theorem gives the remainder on dividing  $p(x)$  by  $(x - a)$  as  $p(a)$ , and the factor theorem says  $(x - a)$  is a factor exactly when  $p(a) = 0$ .

**Real-Life Example**

Checking whether a large polynomial has a particular factor takes one substitution instead of a full long division — the theorem turns pages of work into a single line.

**Detailed Explanation****Definition**

Remainder theorem: if a polynomial  $p(x)$  is divided by the linear polynomial  $(x - a)$ , the remainder is  $p(a)$ . Factor theorem:  $(x - a)$  is a factor of  $p(x)$  if and only if  $p(a) = 0$ .

### Key Idea

The factor theorem is simply the remainder theorem with remainder zero. Recognising this means only one idea has to be remembered, not two.

### Working Principle

To factorise a cubic, guess a small integer zero from the factors of the constant term, confirm with the factor theorem, divide out the linear factor, and factorise the remaining quadratic.

### Applications

- To divide by  $(ax + b)$ , evaluate  $p(-b/a)$ , not  $p(b/a)$ .
- Factorising  $x^3 - 3x^2 - x + 3$  begins by testing  $x = 1$ , which gives zero.
- Splitting the middle term factorises a quadratic without any division.

#### ✓ Points to Remember (Exam Focus)

- ✓ Remainder on division by  $(x - a)$  is  $p(a)$ .
- ✓  $(x - a)$  is a factor of  $p(x)$  if and only if  $p(a) = 0$ .
- ✓ For divisor  $(ax + b)$ , the relevant value is  $x = -b/a$ .
- ✓ Try factors of the constant term first when hunting for a zero.
- ✓ Splitting the middle term: find two numbers whose product is  $ac$  and sum is  $b$ .

#### ⚠ Common Student Mistakes

- ⚠ Substituting  $x = a$  when dividing by  $(x + a)$  — the correct value is  $x = -a$ .
- ⚠ Using  $p(b/a)$  instead of  $p(-b/a)$  for the divisor  $(ax + b)$ .
- ⚠ Forgetting to factorise the quotient completely after removing one factor.

#### Memory Hook

'Set the divisor to zero, substitute, and read the remainder.'

#### Advanced Insights

Synthetic division carries out the same computation with only the coefficients, and is far faster for repeated testing. It is the algorithm behind most computer routines for polynomial root-finding.

## Formula Section

### Remainder Theorem

|                                |                                                                                    |
|--------------------------------|------------------------------------------------------------------------------------|
| <b>Difficulty</b>              | ● Easy ★☆☆                                                                         |
| <b>Formula</b>                 | Remainder of $p(x) \div (x - a) = p(a)$                                            |
| <b>Meaning of Variables</b>    | $p(x)$ = the polynomial, $a$ = the value that makes the divisor zero.              |
| <b>When to Use</b>             | To find a remainder without performing long division.                              |
| <b>Common Mistakes</b>         | Getting the sign of $a$ wrong for a divisor of the form $(x + a)$ .                |
| <b>Shortcut / Memory Trick</b> | Set the divisor equal to zero and solve for $x$ — that is the value to substitute. |

### Algebraic Identities

|                                |                                                                                                                    |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Difficulty</b>              | ● Medium ★★☆☆                                                                                                      |
| <b>Formula</b>                 | $(a+b)^2 = a^2+2ab+b^2$ ; $a^2-b^2 = (a+b)(a-b)$ ; $(a+b)^3 = a^3+3a^2b+3ab^2+b^3$ ; $a^3+b^3 = (a+b)(a^2-ab+b^2)$ |
| <b>Meaning of Variables</b>    | $a$ and $b$ = any algebraic terms.                                                                                 |
| <b>When to Use</b>             | For rapid expansion and factorisation without long multiplication.                                                 |
| <b>Common Mistakes</b>         | Writing $(a+b)^2 = a^2 + b^2$ and omitting the middle term.                                                        |
| <b>Shortcut / Memory Trick</b> | $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$ ; if $a+b+c = 0$ then $a^3+b^3+c^3 = 3abc$ .              |

## II Quick Recap — Topics Covered So Far

- **Polynomials, Degree and Zeroes** — A polynomial is an algebraic expression in which the powers of the variable are whole numbers, and a zero of a polynomial is a value of the variable that makes it equal to zero.
- **Remainder Theorem and Factor Theorem** — The remainder theorem gives the remainder on dividing  $p(x)$  by  $(x - a)$  as  $p(a)$ , and the factor theorem says  $(x - a)$  is a factor exactly when  $p(a) = 0$ .

## Chapter 2 — End of Chapter Revision

### All Memory Hooks

- Polynomials, Degree and Zeroes: 'Whole-number powers only' — that one rule decides whether an expression is a polynomial.
- Remainder Theorem and Factor Theorem: 'Set the divisor to zero, substitute, and read the remainder.'

### All Common Mistakes

-  Calling  $\sqrt{x} + 3$  or  $1/x + 2$  a polynomial; fractional and negative powers are not allowed.
-  Saying the degree of a non-zero constant is undefined; it is 0.
-  Confusing the value of a polynomial  $p(k)$  with a zero of the polynomial.
-  Substituting  $x = a$  when dividing by  $(x + a)$  — the correct value is  $x = -a$ .
-  Using  $p(b/a)$  instead of  $p(-b/a)$  for the divisor  $(ax + b)$ .
-  Forgetting to factorise the quotient completely after removing one factor.

### Formula Sheet

| Formula                                                                                                                                         | Meaning / Use                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Remainder Theorem:</b> Remainder of $p(x) \div (x - a) = p(a)$                                                                               | To find a remainder without performing long division.              |
| <b>Algebraic Identities:</b> $(a+b)^2 = a^2+2ab+b^2$ ; $a^2-b^2 = (a+b)(a-b)$ ; $(a+b)^3 = a^3+3a^2b+3ab^2+b^3$ ; $a^3+b^3 = (a+b)(a^2-ab+b^2)$ | For rapid expansion and factorisation without long multiplication. |

### Quick Revision Section

- Polynomials, Degree and Zeroes: A polynomial is an algebraic expression in which the powers of the variable are whole numbers, and a zero of a polynomial is a value of the variable that makes it equal to zero.
- Remainder Theorem and Factor Theorem: The remainder theorem gives the remainder on dividing  $p(x)$  by  $(x - a)$  as  $p(a)$ , and the factor theorem says  $(x - a)$  is a factor exactly when  $p(a) = 0$ .

## Chapter 3: Co-ordinate Geometry

### Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Ability to plot points on a number line.
- Understanding of positive and negative directions.
- Familiarity with the terms horizontal and vertical.
- Basic knowledge of the Pythagoras theorem.

### 3.1 The Cartesian Plane and Plotting Points

 **Estimated time:** 14 mins   **Difficulty:**  Easy        **Exam Importance:**     

#### Quick Summary

Two perpendicular number lines divide the plane into four quadrants, and every point is located by an ordered pair  $(x, y)$  giving its distances from the two axes.

#### Real-Life Example

A map reference, a seat number in a cinema and a pixel address on a screen are all Cartesian coordinates under different names.

#### Detailed Explanation

##### Definition

The x-axis and y-axis intersect at right angles at the origin  $O(0, 0)$ . The abscissa is the x-coordinate (perpendicular distance from the y-axis) and the ordinate is the y-coordinate (perpendicular distance from the x-axis).

##### Key Idea

The order matters absolutely:  $(3, 5)$  and  $(5, 3)$  are different points. This is why  $(x, y)$  is called an ordered pair.

##### Working Principle

Signs of the coordinates identify the quadrant. Points on the x-axis have  $y = 0$ , and points on the y-axis have  $x = 0$  — which is how axis membership is tested.

##### Applications

- Quadrant I  $(+, +)$ , II  $(-, +)$ , III  $(-, -)$ , IV  $(+, -)$ , counted anticlockwise.
- Graphs of linear equations are drawn by plotting and joining points.
- Computer graphics addresses every pixel by a coordinate pair.

✓ **Points to Remember (Exam Focus)**

- ✓ The origin is (0, 0); the axes divide the plane into four quadrants.
- ✓ Abscissa = x-coordinate; ordinate = y-coordinate.
- ✓ Quadrant signs: I (+,+), II (-,+), III (-,-), IV (+,-).
- ✓ A point on the x-axis is (a, 0); a point on the y-axis is (0, b).
- ✓ (x, y) is an ordered pair — reversing it gives a different point.

### ⚠ Common Student Mistakes

- ⚠ Plotting (5, 3) where (3, 5) was intended.
- ⚠ Numbering the quadrants clockwise instead of anticlockwise.
- ⚠ Saying a point on the x-axis has  $x = 0$ ; it is  $y$  that is zero.



### Memory Hook

'x first, then y — walk before you climb.'



### Advanced Insights

Descartes' insight in uniting algebra and geometry meant every geometric statement could become an equation. Coordinate geometry is the bridge that makes calculus, physics and computer graphics possible.

## Formula Section



### Distance Formula

|                         |                                                                                              |
|-------------------------|----------------------------------------------------------------------------------------------|
| Difficulty              | ● Medium ★★☆☆                                                                                |
| Formula                 | $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$                                                   |
| Meaning of Variables    | $(x_1, y_1)$ and $(x_2, y_2)$ = the two points, $d$ = distance between them.                 |
| When to Use             | To find the length of a segment, or to test whether a triangle is isosceles or right-angled. |
| Common Mistakes         | Forgetting to square the differences, or subtracting the coordinates in a mixed order.       |
| Shortcut / Memory Trick | Distance from the origin to $(x, y)$ is simply $\sqrt{x^2 + y^2}$ .                          |

## II Quick Recap — Topics Covered So Far

- The Cartesian Plane and Plotting Points — Two perpendicular number lines divide the plane into four quadrants, and every point is located by an ordered pair  $(x, y)$  giving its distances from the two axes.

## Chapter 3 — End of Chapter Revision

### All Memory Hooks

- The Cartesian Plane and Plotting Points: 'x first, then y — walk before you climb.'

### All Common Mistakes

-  Plotting (5, 3) where (3, 5) was intended.
-  Numbering the quadrants clockwise instead of anticlockwise.
-  Saying a point on the x-axis has x = 0; it is y that is zero.

### Formula Sheet

| Formula                                                             | Meaning / Use                                                                                |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Distance Formula:</b> $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ | To find the length of a segment, or to test whether a triangle is isosceles or right-angled. |

### Quick Revision Section

- The Cartesian Plane and Plotting Points: Two perpendicular number lines divide the plane into four quadrants, and every point is located by an ordered pair (x, y) giving its distances from the two axes.

## Chapter 4: Linear Equations in Two Variables

### Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Ability to solve a linear equation in one variable.
- Familiarity with the Cartesian plane and plotting points.
- Understanding of substitution into an algebraic expression.
- Knowledge of the terms coefficient and constant.

### 4.1 Linear Equations and Their Solutions

 **Estimated time:** 14 mins   **Difficulty:**  Medium   **★★☆**   **Exam Importance:**     

#### Quick Summary

An equation of the form  $ax + by + c = 0$  with  $a$  and  $b$  not both zero is linear in two variables, and it has infinitely many solutions.

#### Real-Life Example

If two pens and three pencils cost ₹50, there are many possible price combinations — a single equation in two unknowns can never pin down both.

#### Detailed Explanation

##### Definition

A linear equation in two variables is an equation of the form  $ax + by + c = 0$ , where  $a$ ,  $b$  and  $c$  are real numbers and  $a$  and  $b$  are not both zero. A solution is an ordered pair  $(x, y)$  that satisfies the equation.

##### Key Idea

One equation in two variables has infinitely many solutions because each chosen value of  $x$  determines a value of  $y$ . Only a second independent equation can fix a unique pair.

##### Working Principle

To find solutions, substitute any convenient value for one variable and solve for the other. Choosing  $x = 0$  and  $y = 0$  gives the two intercepts, which are usually the easiest points to plot.

##### Applications

- The graph of a linear equation in two variables is always a straight line.
- Every point on the line is a solution, and every solution lies on the line.
- $y = mx$  is a line through the origin;  $x = a$  is vertical and  $y = b$  is horizontal.

✓ **Points to Remember (Exam Focus)**

- ✓ Standard form:  $ax + by + c = 0$ , with  $a$  and  $b$  not both zero.
- ✓ A linear equation in two variables has infinitely many solutions.
- ✓ Its graph is always a straight line; two points are enough to draw it.
- ✓  $x = a$  is a line parallel to the  $y$ -axis;  $y = b$  is parallel to the  $x$ -axis.
- ✓ An equation in one variable represents a point on the number line but a line in the plane.

### ⚠ Common Student Mistakes

- ⚠ Saying a linear equation in two variables has only one solution.
- ⚠ Plotting only one point and trying to draw the line through it.
- ⚠ Confusing  $x = 3$  (a vertical line in the plane) with the single point 3 on the number line.

### 🧠 Memory Hook

**'One equation, two unknowns, infinitely many answers — and all of them lie on one line.'**

### 🚀 Advanced Insights

The line is the set of all solutions, so solving simultaneous equations is geometrically the intersection of two lines. Parallel lines mean no solution and coincident lines mean infinitely many — algebra and geometry saying the same thing.

## II Quick Recap — Topics Covered So Far

- Linear Equations and Their Solutions — An equation of the form  $ax + by + c = 0$  with  $a$  and  $b$  not both zero is linear in two variables, and it has infinitely many solutions.

## Chapter 4 — End of Chapter Revision

### All Memory Hooks

- Linear Equations and Their Solutions: 'One equation, two unknowns, infinitely many answers — and all of them lie on one line.'

### All Common Mistakes

-  Saying a linear equation in two variables has only one solution.
-  Plotting only one point and trying to draw the line through it.
-  Confusing  $x = 3$  (a vertical line in the plane) with the single point 3 on the number line.

### Quick Revision Section

- Linear Equations and Their Solutions: An equation of the form  $ax + by + c = 0$  with  $a$  and  $b$  not both zero is linear in two variables, and it has infinitely many solutions.

## Chapter 5: Introduction to Euclid's Geometry

### Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Familiarity with points, lines and planes.
- Idea that a statement can be proved from simpler statements.
- Basic knowledge of angles and triangles.
- Understanding of the words definition and assumption.

### 5.1 Euclid's Axioms and Postulates

 **Estimated time:** 14 mins   **Difficulty:**  Medium   **★★☆**   **Exam Importance:**    

#### Quick Summary

Euclid built the whole of geometry from a small number of undefined terms, self-evident axioms and five geometrical postulates.

#### Real-Life Example

Every legal system works the same way — a few agreed principles, from which every specific ruling is derived by argument.

### Detailed Explanation

#### Definition

Axioms are assumptions used throughout mathematics and are not specific to geometry. Postulates are the assumptions specific to geometry. A theorem is a statement proved from axioms, postulates and previously proved theorems.

#### Key Idea

Some terms — point, line, plane — must be left undefined, because defining everything would require an endless chain. Euclid accepted this and started from undefined primitives.

#### Working Principle

Euclid's fifth postulate, about lines meeting on the side where interior angles sum to less than two right angles, is equivalent to the parallel postulate, and is far less self-evident than the other four.

#### Applications

- Axiom: things equal to the same thing are equal to one another.
- Postulate: a straight line may be drawn from any point to any other point.
- Postulate: all right angles are equal to one another.

### ✓ Points to Remember (Exam Focus)

- ✓ Axioms are general assumptions; postulates are geometry-specific assumptions.
- ✓ Point, line and plane are undefined terms.
- ✓ Euclid gave five postulates and a number of common notions (axioms).
- ✓ The fifth postulate is equivalent to Playfair's axiom on unique parallels.
- ✓ A theorem must be proved; an axiom or postulate is assumed.

### ⚠ Common Student Mistakes

- ⚠ Using axiom and postulate interchangeably; postulates are specific to geometry.
- ⚠ Trying to define a point or a line; these are deliberately undefined.
- ⚠ Assuming Euclid proved the fifth postulate; he assumed it, and centuries of attempts to prove it failed.



### Memory Hook

**'Axioms are for all mathematics, Postulates are for geometry, Theorems must be earned.'**



### Advanced Insights

Denying the fifth postulate does not produce a contradiction — it produces non-Euclidean geometry. Those geometries turned out to describe the actual universe in Einstein's general relativity.

## II Quick Recap — Topics Covered So Far

- Euclid's Axioms and Postulates — Euclid built the whole of geometry from a small number of undefined terms, self-evident axioms and five geometrical postulates.

## Chapter 5 — End of Chapter Revision

### All Memory Hooks

- Euclid's Axioms and Postulates: 'Axioms are for all mathematics, Postulates are for geometry, Theorems must be earned.'

### All Common Mistakes

-  Using axiom and postulate interchangeably; postulates are specific to geometry.
-  Trying to define a point or a line; these are deliberately undefined.
-  Assuming Euclid proved the fifth postulate; he assumed it, and centuries of attempts to prove it failed.

### Quick Revision Section

- Euclid's Axioms and Postulates: Euclid built the whole of geometry from a small number of undefined terms, self-evident axioms and five geometrical postulates.

## Chapter 6: Lines and Angles

### Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Knowledge of types of angles — acute, right, obtuse, straight.
- Ability to measure an angle with a protractor.
- Idea of parallel and intersecting lines.
- Understanding that angles on a straight line sum to  $180^\circ$ .

### 6.1 Pairs of Angles and Intersecting Lines

 **Estimated time:** 14 mins   **Difficulty:**  Easy ★☆☆   **Exam Importance:** ★★★★★

#### Quick Summary

Angles formed at a point or on a line follow fixed relationships — linear pairs sum to  $180^\circ$  and vertically opposite angles are always equal.

#### Real-Life Example

Two crossing roads always create equal angles opposite each other, which is why a junction looks symmetric from opposite corners.

### Detailed Explanation

#### Definition

Complementary angles sum to  $90^\circ$ ; supplementary angles sum to  $180^\circ$ . A linear pair is a pair of adjacent angles whose non-common arms form a straight line. Vertically opposite angles are the pairs of opposite angles formed by two intersecting lines.

#### Key Idea

Vertically opposite angles are equal because each is supplementary to the same adjacent angle. The proof is one line, and it is a standard two-mark question.

#### Working Principle

The sum of all angles at a point is  $360^\circ$ , and the sum of angles on one side of a straight line is  $180^\circ$ . Every angle-chasing problem reduces to repeated use of these two facts.

#### Applications

- Adjacent angles share a common vertex and a common arm but no common interior points.
- If two lines intersect, both pairs of vertically opposite angles are equal.
- Angles around a point add to  $360^\circ$ .

**✓ Points to Remember (Exam Focus)**

- ✓ Complementary: sum  $90^\circ$ ; Supplementary: sum  $180^\circ$ .
- ✓ A linear pair is always supplementary.
- ✓ Vertically opposite angles are always equal.
- ✓ Angles at a point sum to  $360^\circ$ .
- ✓ Two supplementary adjacent angles form a linear pair, so their outer arms are collinear.

**⚠ Common Student Mistakes**

- ⚠ Swapping complementary ( $90^\circ$ ) and supplementary ( $180^\circ$ ).
- ⚠ Assuming any two equal angles are vertically opposite; they must be formed by two intersecting lines.
- ⚠ Treating adjacent angles as necessarily forming a linear pair.

**Memory Hook**

'C comes before S, and 90 comes before 180' — Complementary 90, Supplementary 180.

**Advanced Insights**

The linear pair axiom and its converse together are equivalent to the statement that a straight angle measures exactly  $180^\circ$ . Much of elementary geometry rests on this single normalisation.

**6.2 Parallel Lines and a Transversal**

🕒 **Estimated time:** 15 mins   **Difficulty:** ● Medium   ★★☆☆ **Exam Importance:** ★★★★★

**Quick Summary**

When a transversal cuts two parallel lines, corresponding angles are equal, alternate angles are equal, and co-interior angles are supplementary.

**Real-Life Example**

Railway sleepers cut across two parallel rails at the same angle every time — that constancy is exactly the corresponding-angles property.

**Detailed Explanation****Definition**

A transversal is a line that intersects two or more lines at distinct points. It creates corresponding, alternate interior, alternate exterior and co-interior (allied) angle pairs.

**Key Idea**

Each of these three properties is both a test for parallelism and a consequence of it. Given parallel lines you may use the equalities; given the equalities you may conclude the lines are parallel.

**Working Principle**

In angle chasing, mark all equal angles first using corresponding and alternate pairs, then use the linear pair and angle-sum facts to fill in the rest.

**Applications**

- Corresponding angles equal  $\Leftrightarrow$  lines are parallel.
- Alternate interior angles equal  $\Leftrightarrow$  lines are parallel.
- Co-interior angles supplementary  $\Leftrightarrow$  lines are parallel.

**✓ Points to Remember (Exam Focus)**

- ✓ Corresponding angles: equal when the lines are parallel (F-shape).
- ✓ Alternate interior angles: equal when the lines are parallel (Z-shape).
- ✓ Co-interior (allied) angles: supplementary when the lines are parallel (C or U-shape).
- ✓ Lines parallel to the same line are parallel to each other.
- ✓ Each property works in both directions — as a consequence and as a test.

**⚠ Common Student Mistakes**

- ⚠ Saying co-interior angles are equal; they are supplementary.
- ⚠ Applying these properties when the lines are not known to be parallel.
- ⚠ Mixing up alternate interior with alternate exterior angles.

**🧠 Memory Hook**

'F for corresponding, Z for alternate, C for co-interior' — and only C adds to  $180^\circ$ .

**🚀 Advanced Insights**

All three properties depend on Euclid's fifth postulate. On a sphere, where 'lines' are great circles, no two lines are parallel at all and none of these results holds.

**II Quick Recap — Topics Covered So Far**

- Pairs of Angles and Intersecting Lines — Angles formed at a point or on a line follow fixed relationships — linear pairs sum to  $180^\circ$  and vertically opposite angles are always equal.

- **Parallel Lines and a Transversal** — When a transversal cuts two parallel lines, corresponding angles are equal, alternate angles are equal, and co-interior angles are supplementary.

## 6.3 Angle Sum Property of a Triangle

**Estimated time:** 13 mins   **Difficulty:** Medium   **Exam Importance:** ★★☆☆

### Quick Summary

The three angles of a triangle always add to  $180^\circ$ , and an exterior angle equals the sum of the two interior opposite angles.

### Real-Life Example

A surveyor checks measurements by confirming that the three measured angles of a triangular plot total  $180^\circ$  — any excess signals an error.

### Detailed Explanation

#### Definition

The angle sum property states that the sum of the three interior angles of a triangle is  $180^\circ$ . The exterior angle property states that an exterior angle equals the sum of the two interior opposite angles.

#### Key Idea

The exterior angle property follows immediately from the angle sum property together with the linear pair at that vertex, so the second result is really a corollary of the first.

#### Working Principle

The angle sum is proved by drawing a line through one vertex parallel to the opposite side and using alternate angles. This is the standard proof expected in the examination.

#### Applications

- A triangle can have at most one right angle or one obtuse angle.
- In an equilateral triangle each angle is  $60^\circ$ .
- The exterior angle is always greater than either interior opposite angle.

#### ✓ Points to Remember (Exam Focus)

- ✓ Sum of the interior angles of a triangle =  $180^\circ$ .
- ✓ Exterior angle = sum of the two interior opposite angles.
- ✓ A triangle cannot have two right angles or two obtuse angles.
- ✓ The angle sum property is proved using a parallel line through one vertex.
- ✓ Sum of the exterior angles of any polygon, taken one at each vertex, is  $360^\circ$ .

### Common Student Mistakes

-  Adding the exterior angle to all three interior angles.
-  Assuming the exterior angle equals the adjacent interior angle.
-  Forgetting that the property holds only in plane (Euclidean) geometry.

### Memory Hook

**'Three angles, one straight line —  $180^\circ$  every time.'**

### Advanced Insights

On a sphere the angles of a triangle sum to more than  $180^\circ$ , and the excess is proportional to its area. Gauss reportedly tried to measure this excess on mountain triangles to test whether physical space is Euclidean.

## II Quick Recap — Topics Covered So Far

- Angle Sum Property of a Triangle — The three angles of a triangle always add to  $180^\circ$ , and an exterior angle equals the sum of the two interior opposite angles.

## Chapter 6 — End of Chapter Revision

### All Memory Hooks

- Pairs of Angles and Intersecting Lines: 'C comes before S, and 90 comes before 180' — Complementary 90, Supplementary 180.
- Parallel Lines and a Transversal: 'F for corresponding, Z for alternate, C for co-interior' — and only C adds to  $180^\circ$ .
- Angle Sum Property of a Triangle: 'Three angles, one straight line —  $180^\circ$  every time.'

### All Common Mistakes

-  Swapping complementary ( $90^\circ$ ) and supplementary ( $180^\circ$ ).
-  Assuming any two equal angles are vertically opposite; they must be formed by two intersecting lines.
-  Treating adjacent angles as necessarily forming a linear pair.
-  Saying co-interior angles are equal; they are supplementary.
-  Applying these properties when the lines are not known to be parallel.
-  Mixing up alternate interior with alternate exterior angles.
-  Adding the exterior angle to all three interior angles.
-  Assuming the exterior angle equals the adjacent interior angle.
-  Forgetting that the property holds only in plane (Euclidean) geometry.

### Quick Revision Section

- Pairs of Angles and Intersecting Lines: Angles formed at a point or on a line follow fixed relationships — linear pairs sum to  $180^\circ$  and vertically opposite angles are always equal.
- Parallel Lines and a Transversal: When a transversal cuts two parallel lines, corresponding angles are equal, alternate angles are equal, and co-interior angles are supplementary.
- Angle Sum Property of a Triangle: The three angles of a triangle always add to  $180^\circ$ , and an exterior angle equals the sum of the two interior opposite angles.

## Chapter 7: Triangles

### Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Angle sum property of a triangle.
- Properties of parallel lines and a transversal.
- Idea of congruent figures as identical in shape and size.
- Familiarity with the terms median, altitude and bisector.

### 7.1 Congruence of Triangles

 **Estimated time:** 16 mins   **Difficulty:**  Medium   **★★☆**   **Exam Importance:**     

#### Quick Summary

Two triangles are congruent if they have exactly the same shape and size, and this can be established by the SSS, SAS, ASA, AAS or RHS criteria.

#### Real-Life Example

Mass-produced roof trusses must be congruent so that any one fits any position — manufacturers verify this by checking three measurements, not six.

#### Detailed Explanation

##### Definition

Two triangles are congruent if their corresponding sides and corresponding angles are equal. The congruence criteria are shortcuts that establish this from only three pieces of information.

##### Key Idea

SAS requires the angle to be between the two sides. SSA is not a valid criterion — two different triangles can share two sides and a non-included angle, which is why it is excluded.

##### Working Principle

Write the correspondence in the correct order:  $\triangle ABC \cong \triangle PQR$  means  $A \leftrightarrow P$ ,  $B \leftrightarrow Q$ ,  $C \leftrightarrow R$ . All conclusions about corresponding parts (CPCT) depend on getting this order right.

##### Applications

- RHS applies only to right triangles, using the hypotenuse and one side.
- AAA is not a congruence criterion — it gives similarity, not congruence.
- CPCT is used after proving congruence to deduce equality of remaining parts.

✓ **Points to Remember (Exam Focus)**

- ✓ Congruence criteria: SSS, SAS, ASA, AAS and RHS.
- ✓ SAS requires the INCLUDED angle between the two sides.
- ✓ SSA and AAA are NOT congruence criteria.
- ✓ Write the correspondence in the correct order when stating congruence.
- ✓ CPCT — Corresponding Parts of Congruent Triangles are equal — is used after congruence is proved.

### ⚠ Common Student Mistakes

- ⚠ Using SSA as a congruence criterion.
- ⚠ Writing the vertices in the wrong order in a congruence statement.
- ⚠ Applying AAA and concluding congruence rather than similarity.

### 🧠 Memory Hook

'SSS, SAS, ASA, AAS, RHS' — five valid keys; SSA and AAA open nothing.

### 🚀 Advanced Insights

SSA fails because a circle of given radius can cut a line in two places, giving the ambiguous case. In trigonometry this reappears as the two possible solutions of the sine rule.

## 7.2 Properties of Triangles and Inequalities

🕒 **Estimated time:** 15 mins   **Difficulty:** ● Medium   ★★☆☆ **Exam Importance:** ★★★★★

### 🔑 Quick Summary

Angles opposite equal sides are equal, the larger angle lies opposite the larger side, and any two sides together exceed the third.

### 🌍 Real-Life Example

You cannot build a triangle from sticks of 2 cm, 3 cm and 8 cm — the two shorter sticks simply will not reach across.

### 📖 Detailed Explanation

#### Definition

The triangle inequality states that the sum of the lengths of any two sides of a triangle is greater than the third side. In an isosceles triangle, the angles opposite the equal sides are equal.

**Key Idea**

Side–angle relationships always run in the same direction: bigger side, bigger opposite angle. This one principle covers both the isosceles theorem and the inequality theorems.

**Working Principle**

To test whether three lengths can form a triangle, it is enough to check that the two smaller lengths together exceed the largest. The other two inequalities then follow automatically.

**Applications**

- Angles opposite equal sides are equal, and conversely.
- The side opposite the greater angle is longer.
- The difference of any two sides is less than the third side.

**✓ Points to Remember (Exam Focus)**

- ✓ Isosceles triangle: angles opposite the equal sides are equal (and conversely).
- ✓ The greater side has the greater angle opposite it.
- ✓ Sum of any two sides  $>$  the third side.
- ✓ Difference of any two sides  $<$  the third side.
- ✓ The perpendicular from a point to a line is the shortest distance to that line.

**⚠ Common Student Mistakes**

- ⚠ Checking only one of the three triangle inequalities when a length is large.
- ⚠ Reversing the side–angle relation and pairing the largest side with the smallest angle.
- ⚠ Assuming an isosceles triangle must also be acute-angled.

**💡 Memory Hook**

**'Bigger side, bigger angle' — and 'the two short ones must out-reach the long one'.**

**🚀 Advanced Insights**

The triangle inequality generalises far beyond geometry: it is one of the defining axioms of a metric space, and holds for distances in any dimension, for absolute values, and for vector magnitudes.

**II Quick Recap — Topics Covered So Far**

- Congruence of Triangles — Two triangles are congruent if they have exactly the same shape and size, and this can be established by the SSS, SAS, ASA, AAS or RHS criteria.

- Properties of Triangles and Inequalities — Angles opposite equal sides are equal, the larger angle lies opposite the larger side, and any two sides together exceed the third.

## Chapter 7 — End of Chapter Revision

### All Memory Hooks

- Congruence of Triangles: 'SSS, SAS, ASA, AAS, RHS' — five valid keys; SSA and AAA open nothing.
- Properties of Triangles and Inequalities: 'Bigger side, bigger angle' — and 'the two short ones must out-reach the long one'.

### All Common Mistakes

-  Using SSA as a congruence criterion.
-  Writing the vertices in the wrong order in a congruence statement.
-  Applying AAA and concluding congruence rather than similarity.
-  Checking only one of the three triangle inequalities when a length is large.
-  Reversing the side–angle relation and pairing the largest side with the smallest angle.
-  Assuming an isosceles triangle must also be acute-angled.

### Quick Revision Section

- Congruence of Triangles: Two triangles are congruent if they have exactly the same shape and size, and this can be established by the SSS, SAS, ASA, AAS or RHS criteria.
- Properties of Triangles and Inequalities: Angles opposite equal sides are equal, the larger angle lies opposite the larger side, and any two sides together exceed the third.

## Chapter 8: Quadrilaterals

### Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Angle sum property of a triangle.
- Congruence criteria for triangles.
- Properties of parallel lines and a transversal.
- Familiarity with the terms diagonal and vertex.

### 8.1 Types of Quadrilaterals and the Parallelogram

 **Estimated time:** 15 mins   **Difficulty:**  Medium   **★★☆**   **Exam Importance:**     

#### Quick Summary

The angles of a quadrilateral sum to  $360^\circ$ , and a parallelogram has opposite sides and angles equal with diagonals that bisect each other.

#### Real-Life Example

A collapsible scissor gate stays a parallelogram at every stage of opening — opposite sides remain equal even as the angles change.

#### Detailed Explanation

##### Definition

A parallelogram is a quadrilateral in which both pairs of opposite sides are parallel. A rhombus is a parallelogram with all sides equal, a rectangle one with all angles right angles, and a square is both.

##### Key Idea

The family is hierarchical: every square is a rhombus and a rectangle, and every rhombus and rectangle is a parallelogram. Properties inherit downward but not upward.

##### Working Principle

A quadrilateral is a parallelogram if any one of these holds: both pairs of opposite sides equal, both pairs of opposite angles equal, diagonals bisect each other, or one pair of opposite sides both equal and parallel.

##### Applications

- Diagonals of a rectangle are equal; of a rhombus, perpendicular; of a square, both.
- A diagonal divides a parallelogram into two congruent triangles.
- The angle sum of any quadrilateral is  $360^\circ$ .

✓ **Points to Remember (Exam Focus)**

- ✓ Angle sum of a quadrilateral =  $360^\circ$ .
- ✓ Parallelogram: opposite sides equal, opposite angles equal, diagonals bisect each other.
- ✓ Rectangle: diagonals are equal. Rhombus: diagonals are perpendicular bisectors.
- ✓ Square: diagonals are equal AND perpendicular bisectors.
- ✓ One pair of opposite sides equal and parallel is enough to prove a parallelogram.

### ⚠ Common Student Mistakes

- ⚠ Saying the diagonals of a parallelogram are equal; only in a rectangle are they equal.
- ⚠ Assuming the diagonals of a rhombus are equal; they are perpendicular, not equal.
- ⚠ Calling a trapezium a parallelogram; only one pair of its sides is parallel.

### 🧠 Memory Hook

'Square = Rectangle + Rhombus' — it inherits equal diagonals from one and perpendicular diagonals from the other.

### 🚀 Advanced Insights

In vector terms, a quadrilateral ABCD is a parallelogram exactly when  $\vec{AB} = \vec{DC}$ . Every classical property then follows in one or two lines of vector algebra.

## 8.2 The Mid-point Theorem

🕒 **Estimated time:** 13 mins   **Difficulty:** ● Medium   ★★☆☆ **Exam Importance:** ★★★★★

### 🔑 Quick Summary

The line joining the mid-points of two sides of a triangle is parallel to the third side and half its length.

### 🌍 Real-Life Example

A shelf fixed halfway up two sloping supports is automatically parallel to the floor and half the span — the mid-point theorem doing carpentry.

### 📖 Detailed Explanation

#### Definition

Mid-point theorem: the segment joining the mid-points of two sides of a triangle is parallel to the third side and equal to half of it. Its converse states that a line through the mid-point of one side, parallel to another side, bisects the third side.

**Key Idea**

The theorem converts a statement about lengths into one about parallels, and its converse does the reverse. Most quadrilateral proofs use one or the other to create a parallelogram.

**Working Principle**

In a quadrilateral, joining the mid-points of all four sides always produces a parallelogram, whatever the original shape. This is proved by applying the mid-point theorem to the two triangles formed by a diagonal.

**Applications**

- The figure formed by joining the mid-points of any quadrilateral is a parallelogram.
- Mid-points of a rectangle give a rhombus; of a rhombus, a rectangle.
- The converse is used to prove that a point is a mid-point.

**✓ Points to Remember (Exam Focus)**

- ✓ Mid-point theorem: segment joining mid-points is parallel to and half the third side.
- ✓ Converse: a line through one mid-point parallel to another side bisects the third side.
- ✓ Joining mid-points of any quadrilateral gives a parallelogram.
- ✓ Mid-points of a rectangle form a rhombus; mid-points of a rhombus form a rectangle.
- ✓ The theorem is proved using congruent triangles or by constructing a parallelogram.

**⚠ Common Student Mistakes**

- ⚠ Forgetting the 'half the length' part and stating only parallelism.
- ⚠ Applying the converse without confirming the parallel condition.
- ⚠ Assuming the mid-point figure of a square is a square of half the area; it is a square of half the area, but the same is not true for other shapes.

**💡 Memory Hook**

'Join the mid-points: parallel to the third side, and exactly half of it.'

**🚀 Advanced Insights**

The mid-point theorem is the simplest case of the basic proportionality theorem, which allows any ratio rather than just one-half. Class 10 similarity generalises this chapter almost entirely.

**II Quick Recap — Topics Covered So Far**

- Types of Quadrilaterals and the Parallelogram — The angles of a quadrilateral sum to  $360^\circ$ , and a parallelogram has opposite sides and angles equal with diagonals that bisect each other.

- The Mid-point Theorem — The line joining the mid-points of two sides of a triangle is parallel to the third side and half its length.

## Chapter 8 — End of Chapter Revision

### All Memory Hooks

- Types of Quadrilaterals and the Parallelogram: 'Square = Rectangle + Rhombus' — it inherits equal diagonals from one and perpendicular diagonals from the other.
- The Mid-point Theorem: 'Join the mid-points: parallel to the third side, and exactly half of it.'

### All Common Mistakes

-  Saying the diagonals of a parallelogram are equal; only in a rectangle are they equal.
-  Assuming the diagonals of a rhombus are equal; they are perpendicular, not equal.
-  Calling a trapezium a parallelogram; only one pair of its sides is parallel.
-  Forgetting the 'half the length' part and stating only parallelism.
-  Applying the converse without confirming the parallel condition.
-  Assuming the mid-point figure of a square is a square of half the area; it is a square of half the area, but the same is not true for other shapes.

### Quick Revision Section

- Types of Quadrilaterals and the Parallelogram: The angles of a quadrilateral sum to  $360^\circ$ , and a parallelogram has opposite sides and angles equal with diagonals that bisect each other.
- The Mid-point Theorem: The line joining the mid-points of two sides of a triangle is parallel to the third side and half its length.

## Chapter 9: Area of Parallelograms and Triangles

### Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Formula for the area of a rectangle.
- Properties of parallelograms.
- Idea of the height or altitude of a figure.
- Congruence of triangles.

### 9.1 Figures on the Same Base and Between the Same Parallels

 **Estimated time:** 15 mins   **Difficulty:**  Medium   **★★☆**   **Exam Importance:**     

#### Quick Summary

Parallelograms on the same base and between the same parallels are equal in area, and a triangle in that position has half the area of the parallelogram.

#### Real-Life Example

Sliding the top of a stack of paper sideways changes the shape from a rectangle to a slanted parallelogram, but not the amount of paper — area unchanged.

#### Detailed Explanation

##### Definition

Two figures are said to be on the same base and between the same parallels if they share a base and their opposite sides or vertices lie on a line parallel to that base.

##### Key Idea

Area depends only on base and perpendicular height, never on how slanted the figure is. Every result in this chapter is a restatement of that single fact.

##### Working Principle

Since the parallels are a fixed distance apart, all such figures have the same height. With base and height both equal, their areas must be equal.

##### Applications

- A median of a triangle divides it into two triangles of equal area.
- A diagonal of a parallelogram divides it into two congruent triangles of equal area.
- The diagonals of a parallelogram divide it into four triangles of equal area.

✓ **Points to Remember (Exam Focus)**

- ✓ Parallelograms on the same base and between the same parallels are equal in area.
- ✓ A triangle and a parallelogram on the same base and between the same parallels: triangle =  $\frac{1}{2}$  parallelogram.
- ✓ Triangles on the same base and between the same parallels are equal in area.
- ✓ A median divides a triangle into two triangles of equal area.
- ✓ Equal areas do not imply congruence.

### ⚠ Common Student Mistakes

- ⚠ Confusing equal area with congruence; equal-area figures may have very different shapes.
- ⚠ Using a slant side as the height instead of the perpendicular distance.
- ⚠ Forgetting the factor of  $\frac{1}{2}$  when comparing a triangle with a parallelogram.



### Memory Hook

'Same base, same parallels, same area' — slant does not change the size.



### Advanced Insights

This is the geometric ancestor of Cavalieri's principle: two solids with equal cross-sectional areas at every height have equal volume. It is what makes the volume of an oblique cylinder equal to that of a right one.

## Formula Section



### Area of a Parallelogram and a Triangle

|                         |                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------|
| Difficulty              | ● Easy ★☆☆                                                                                                 |
| Formula                 | Area of parallelogram = base $\times$ height; Area of triangle = $\frac{1}{2} \times$ base $\times$ height |
| Meaning of Variables    | base = any chosen side, height = perpendicular distance from that side to the opposite side or vertex.     |
| When to Use             | For any parallelogram or triangle where a base and its corresponding height are known.                     |
| Common Mistakes         | Using a slant side as the height instead of the perpendicular distance.                                    |
| Shortcut / Memory Trick | Any side may be taken as the base, provided the height is measured perpendicular to it.                    |

## II Quick Recap — Topics Covered So Far

- Figures on the Same Base and Between the Same Parallels — Parallelograms on the same base and between the same parallels are equal in area, and a triangle in that position has half the area of the parallelogram.

## Chapter 9 — End of Chapter Revision

### All Memory Hooks

- Figures on the Same Base and Between the Same Parallels: 'Same base, same parallels, same area' — slant does not change the size.

### All Common Mistakes

-  Confusing equal area with congruence; equal-area figures may have very different shapes.
-  Using a slant side as the height instead of the perpendicular distance.
-  Forgetting the factor of  $\frac{1}{2}$  when comparing a triangle with a parallelogram.

### Formula Sheet

| Formula                                                                                                                                                   | Meaning / Use                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>Area of a Parallelogram and a Triangle:</b> Area of parallelogram = base $\times$ height; Area of triangle = $\frac{1}{2} \times$ base $\times$ height | For any parallelogram or triangle where a base and its corresponding height are known. |

### Quick Revision Section

- Figures on the Same Base and Between the Same Parallels: Parallelograms on the same base and between the same parallels are equal in area, and a triangle in that position has half the area of the parallelogram.

## Chapter 10: Circles

### Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Familiarity with the terms radius, diameter and circumference.
- Congruence criteria for triangles.
- Angle sum property of a triangle and of a quadrilateral.
- Idea of a perpendicular from a point to a line.

### 10.1 Chords and Their Properties

 **Estimated time:** 15 mins   **Difficulty:**  Medium   **★★☆**   **Exam Importance:**     

#### Quick Summary

A perpendicular from the centre bisects a chord, equal chords are equidistant from the centre, and the longest chord of a circle is its diameter.

#### Real-Life Example

To find the centre of a circular tabletop, draw two chords and their perpendicular bisectors — where they cross is the centre.

#### Detailed Explanation

##### Definition

A chord is a line segment joining two points on a circle. A diameter is a chord passing through the centre. An arc is a part of the circle, and a segment is the region between a chord and an arc.

##### Key Idea

Every chord property comes from the isosceles triangle formed by the two radii to the chord's endpoints. Once that triangle is drawn, the results follow from congruence.

##### Working Principle

The perpendicular from the centre to a chord bisects it, and conversely the line from the centre to the mid-point of a chord is perpendicular to it. Three non-collinear points determine exactly one circle.

##### Applications

- Equal chords of a circle subtend equal angles at the centre, and conversely.
- Equal chords are equidistant from the centre, and conversely.
- The longer chord is nearer to the centre.

✓ **Points to Remember (Exam Focus)**

- ✓ The perpendicular from the centre to a chord bisects the chord.
- ✓ The line joining the centre to the mid-point of a chord is perpendicular to it.
- ✓ Equal chords are equidistant from the centre, and conversely.
- ✓ The diameter is the longest chord of a circle.
- ✓ One and only one circle passes through three non-collinear points.

### ⚠ Common Student Mistakes

- ⚠ Assuming any line from the centre bisects a chord; it must be perpendicular to it.
- ⚠ Thinking a circle can be drawn through any three points; they must be non-collinear.
- ⚠ Saying the shorter chord is nearer the centre; it is the longer one.

### 🧠 Memory Hook

'Perpendicular from the centre cuts the chord in half' — and 'longer chord, closer to the centre'.

### 🚀 Advanced Insights

The perpendicular-bisector property is what makes trilateration possible. GPS locates a receiver by intersecting spheres, using exactly the three-points-determine-a-circle idea in three dimensions.

## 10.2 Angles in a Circle and Cyclic Quadrilaterals

🕒 **Estimated time:** 15 mins   **Difficulty:** ● Medium   ★★☆☆   **Exam Importance:** ★★★★★

### 🔑 Quick Summary

The angle at the centre is twice the angle at the circumference on the same arc, angles in the same segment are equal, and opposite angles of a cyclic quadrilateral are supplementary.

### 🌐 Real-Life Example

A camera placed anywhere on a certain arc sees a stage of fixed width at exactly the same angle — the equal-angles-in-the-same-segment property.

### 📖 Detailed Explanation

#### Definition

A cyclic quadrilateral is a quadrilateral whose four vertices all lie on a circle. A segment is the region bounded by a chord and one of the arcs it cuts off.

**Key Idea**

The central-angle theorem is the parent result. Angles in the same segment being equal, and the angle in a semicircle being  $90^\circ$ , are both immediate consequences of it.

**Working Principle**

Because the angle at the centre standing on a diameter is  $180^\circ$ , the angle at the circumference on the same arc is  $90^\circ$  — which is why any angle in a semicircle is a right angle.

**Applications**

- The angle in a semicircle is always  $90^\circ$ .
- Opposite angles of a cyclic quadrilateral sum to  $180^\circ$ , and conversely.
- The exterior angle of a cyclic quadrilateral equals the interior opposite angle.

**✓ Points to Remember (Exam Focus)**

- ✓ Angle at the centre =  $2 \times$  angle at the circumference on the same arc.
- ✓ Angles in the same segment of a circle are equal.
- ✓ The angle in a semicircle is a right angle.
- ✓ Opposite angles of a cyclic quadrilateral are supplementary, and conversely.
- ✓ Equal arcs subtend equal angles at the centre.

**⚠ Common Student Mistakes**

- ⚠ Halving instead of doubling when moving from the circumference to the centre.
- ⚠ Applying the equal-angles rule to angles in different segments.
- ⚠ Assuming every quadrilateral is cyclic; the vertices must be concyclic.

**🧠 Memory Hook**

'Centre is double the circumference' — and 'a semicircle always gives a right angle'.

**🚀 Advanced Insights**

The converse of the cyclic quadrilateral theorem is the standard test for concyclic points. Ptolemy's theorem extends it, relating the diagonals and sides of any cyclic quadrilateral in a single elegant equation.

**II Quick Recap — Topics Covered So Far**

- Chords and Their Properties — A perpendicular from the centre bisects a chord, equal chords are equidistant from the centre, and the longest chord of a circle is its diameter.

- Angles in a Circle and Cyclic Quadrilaterals — The angle at the centre is twice the angle at the circumference on the same arc, angles in the same segment are equal, and opposite angles of a cyclic quadrilateral are supplementary.

## Chapter 10 — End of Chapter Revision

### All Memory Hooks

- Chords and Their Properties: 'Perpendicular from the centre cuts the chord in half' — and 'longer chord, closer to the centre'.
- Angles in a Circle and Cyclic Quadrilaterals: 'Centre is double the circumference' — and 'a semicircle always gives a right angle'.

### All Common Mistakes

-  Assuming any line from the centre bisects a chord; it must be perpendicular to it.
-  Thinking a circle can be drawn through any three points; they must be non-collinear.
-  Saying the shorter chord is nearer the centre; it is the longer one.
-  Halving instead of doubling when moving from the circumference to the centre.
-  Applying the equal-angles rule to angles in different segments.
-  Assuming every quadrilateral is cyclic; the vertices must be concyclic.

### Quick Revision Section

- Chords and Their Properties: A perpendicular from the centre bisects a chord, equal chords are equidistant from the centre, and the longest chord of a circle is its diameter.
- Angles in a Circle and Cyclic Quadrilaterals: The angle at the centre is twice the angle at the circumference on the same arc, angles in the same segment are equal, and opposite angles of a cyclic quadrilateral are supplementary.

# Chapter 11: Constructions

## Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Skill in using a ruler and a pair of compasses.
- Properties of triangles and the triangle inequality.
- Knowledge of perpendicular and angle bisectors.
- Angle sum property of a triangle.

## 11.1 Basic Constructions with Ruler and Compasses

 **Estimated time:** 14 mins   **Difficulty:**  Easy   ★☆☆   **Exam Importance:** ★★★★★

### Quick Summary

Using only an ungraduated ruler and compasses, an angle can be bisected, a segment perpendicularly bisected, and angles of  $60^\circ$ ,  $30^\circ$ ,  $90^\circ$  and  $45^\circ$  constructed.

### Real-Life Example

A carpenter marking the exact centre of a plank uses compasses rather than a tape, because the construction is more accurate than any measurement.

### Detailed Explanation

#### Definition

A geometrical construction is an accurate drawing made using only an ungraduated ruler and a pair of compasses, with every step justified by a geometrical theorem.

#### Key Idea

Constructions never rely on measurement. Every step must be justified, which is why arcs must be left visible on the final figure as evidence of the method.

#### Working Principle

An equilateral triangle construction gives  $60^\circ$ ; bisecting it gives  $30^\circ$  and  $15^\circ$ . A perpendicular gives  $90^\circ$ , and bisecting that gives  $45^\circ$  and  $22.5^\circ$ .

#### Applications

- Angles constructible by this method are multiples of  $15^\circ$ .
- The perpendicular bisector of a chord passes through the centre of a circle.
- The point of intersection of two perpendicular bisectors is the circumcentre.

### ✓ Points to Remember (Exam Focus)

- ✓ Only an ungraduated ruler and compasses may be used.
- ✓ Constructible angles:  $60^\circ$ ,  $30^\circ$ ,  $15^\circ$ ,  $90^\circ$ ,  $45^\circ$ ,  $22.5^\circ$ ,  $120^\circ$ ,  $135^\circ$  — all multiples of  $15^\circ$ .
- ✓ Never erase the construction arcs; they are the proof of the method.
- ✓ The perpendicular bisector is the locus of points equidistant from two given points.
- ✓ The angle bisector is the locus of points equidistant from two given lines.

### ⚠ Common Student Mistakes

- ⚠ Measuring with a protractor instead of constructing.
- ⚠ Rubbing out the arcs, which loses the marks for method.
- ⚠ Changing the compass radius midway through a construction that requires it constant.

### 🧠 Memory Hook

'Arcs are the working — never rub them out.'

### 🚀 Advanced Insights

Trisecting an arbitrary angle with ruler and compasses is impossible, proved by Wantzel in 1837 using field theory. A geometry question was finally settled by algebra, more than two thousand years after it was asked.

## 11.2 Construction of Triangles

🕒 **Estimated time:** 14 mins   **Difficulty:** ● Medium   ★★☆☆   **Exam Importance:** ★★★★★

### 🔑 Quick Summary

A triangle can be constructed from a base, a base angle and either the sum or the difference of the other two sides, or from its perimeter and two base angles.

### 🌐 Real-Life Example

A surveyor who knows one side, one angle and the total length of the other two boundaries can still recover the exact shape of a plot.

### 📖 Detailed Explanation

#### Definition

These constructions build a triangle from data that do not directly give all three sides, using an auxiliary point and the isosceles triangle property to convert a sum or difference into a length.

**Key Idea**

The technique is always the same: mark the given sum or difference along the base line, join it, and use the fact that a perpendicular bisector or angle bisector recovers the required vertex.

**Working Principle**

For a given sum of two sides, cut off that length from the base, join the endpoint, and construct the perpendicular bisector of that join. Its intersection with the base line gives the third vertex.

**Applications**

- Given base, base angle and sum of the other two sides.
- Given base, base angle and difference of the other two sides.
- Given perimeter and two base angles.

**✓ Points to Remember (Exam Focus)**

- ✓ Always draw a rough sketch first and mark all the given data on it.
- ✓ The perpendicular bisector method converts a given sum into two equal parts.
- ✓ Construction is possible only if the data satisfy the triangle inequality.
- ✓ The angle sum property limits which pairs of angles can be given.
- ✓ State the steps of construction clearly; marks are awarded for them.

**⚠ Common Student Mistakes**

- ⚠ Attempting a construction whose data violate the triangle inequality.
- ⚠ Omitting the rough sketch, which usually leads to misplacing the given elements.
- ⚠ Not writing the steps of construction when the question asks for them.

**💡 Memory Hook**

'Rough sketch first, then arcs, then steps' — in that order, every time.

**🚀 Advanced Insights**

A construction exists exactly when the corresponding algebraic system has a real solution. The triangle inequality and the angle sum are the geometric expression of that solvability condition.

**II Quick Recap — Topics Covered So Far**

- Basic Constructions with Ruler and Compasses — Using only an ungraduated ruler and compasses, an angle can be bisected, a segment perpendicularly bisected, and angles of  $60^\circ$ ,  $30^\circ$ ,  $90^\circ$  and  $45^\circ$  constructed.

- Construction of Triangles — A triangle can be constructed from a base, a base angle and either the sum or the difference of the other two sides, or from its perimeter and two base angles.

## Chapter 11 — End of Chapter Revision

### All Memory Hooks

- Basic Constructions with Ruler and Compasses: 'Arcs are the working — never rub them out.'
- Construction of Triangles: 'Rough sketch first, then arcs, then steps' — in that order, every time.

### All Common Mistakes

-  Measuring with a protractor instead of constructing.
-  Rubbing out the arcs, which loses the marks for method.
-  Changing the compass radius midway through a construction that requires it constant.
-  Attempting a construction whose data violate the triangle inequality.
-  Omitting the rough sketch, which usually leads to misplacing the given elements.
-  Not writing the steps of construction when the question asks for them.

### Quick Revision Section

- Basic Constructions with Ruler and Compasses: Using only an ungraduated ruler and compasses, an angle can be bisected, a segment perpendicularly bisected, and angles of  $60^\circ$ ,  $30^\circ$ ,  $90^\circ$  and  $45^\circ$  constructed.
- Construction of Triangles: A triangle can be constructed from a base, a base angle and either the sum or the difference of the other two sides, or from its perimeter and two base angles.

## Chapter 12: Heron's Formula

### Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Area of a triangle as  $\frac{1}{2} \times \text{base} \times \text{height}$ .
- Ability to compute square roots.
- Knowledge of the Pythagoras theorem.
- Familiarity with the perimeter of a polygon.

### 12.1 Heron's Formula and Its Applications

 **Estimated time:** 15 mins   **Difficulty:**  Medium   **★★☆**   **Exam Importance:**     

#### Quick Summary

Heron's formula finds the area of a triangle from its three sides alone, without needing the height.

#### Real-Life Example

A farmer with a triangular field can measure the three boundary lengths with a tape and compute the area exactly — no need to find any perpendicular.

#### Detailed Explanation

##### Definition

For a triangle with sides  $a$ ,  $b$  and  $c$  and semi-perimeter  $s = (a + b + c)/2$ , the area is  $\sqrt{s(s-a)(s-b)(s-c)}$ .

##### Key Idea

The formula's whole value is that it needs no height. For a scalene triangle, finding a perpendicular height is awkward, while measuring three sides is easy.

##### Working Principle

Compute  $s$  first, then each of the three differences, then multiply and take the square root. Keeping surds in exact form until the last step avoids accumulated rounding error.

##### Applications

- Area of a quadrilateral: split it along a diagonal into two triangles and add.
- Area of an equilateral triangle of side  $a$  is  $(\sqrt{3}/4)a^2$ .
- Checking whether a triangle is valid: if any  $(s - a)$  is negative, the triangle cannot exist.

#### Points to Remember (Exam Focus)

- ✓  $s = (a + b + c)/2$  is the semi-perimeter, half the perimeter.

- ✓ Area =  $\sqrt{s(s-a)(s-b)(s-c)}$ .
- ✓ Useful when the height is unknown or hard to find.
- ✓ A quadrilateral is handled by splitting it into two triangles along a diagonal.
- ✓ For a right triangle, the ordinary  $\frac{1}{2} \times \text{base} \times \text{height}$  is quicker.

### ⚠ Common Student Mistakes

- ⚠ Using the full perimeter instead of the semi-perimeter.
- ⚠ Forgetting to take the square root at the end.
- ⚠ Rounding intermediate values, which distorts the final answer.

### 🧠 Memory Hook

'Half the perimeter, then s minus each side, multiply all four, square root.'

### 🚀 Advanced Insights

Heron's formula can be derived from the cosine rule with some algebra, and Brahmagupta's formula generalises it to cyclic quadrilaterals. Both are special cases of Bretschneider's formula for any quadrilateral.

## Formula Section

### 📐 Heron's Formula

|                         |                                                                                                        |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Difficulty              | ● Medium ★★☆☆                                                                                          |
| Formula                 | Area = $\sqrt{s(s-a)(s-b)(s-c)}$ , where $s = (a+b+c)/2$                                               |
| Meaning of Variables    | a, b, c = the three sides of the triangle, s = semi-perimeter.                                         |
| When to Use             | Whenever all three sides are known and the height is not.                                              |
| Common Mistakes         | Using the perimeter instead of the semi-perimeter, or omitting the square root.                        |
| Shortcut / Memory Trick | If the triangle is right-angled, $\frac{1}{2} \times \text{base} \times \text{height}$ is far quicker. |

### 📐 Area of an Equilateral Triangle

|                      |                             |
|----------------------|-----------------------------|
| Difficulty           | ● Easy ★☆☆☆                 |
| Formula              | Area = $(\sqrt{3} / 4) a^2$ |
| Meaning of Variables | a = length of each side.    |

|                                |                                                                                               |
|--------------------------------|-----------------------------------------------------------------------------------------------|
| <b>When to Use</b>             | For any equilateral triangle, as a shortcut instead of Heron's formula.                       |
| <b>Common Mistakes</b>         | Using $\sqrt{3}/2$ instead of $\sqrt{3}/4$ , which is the height factor, not the area factor. |
| <b>Shortcut / Memory Trick</b> | Height of an equilateral triangle is $(\sqrt{3}/2)a$ .                                        |

## II Quick Recap — Topics Covered So Far

- Heron's Formula and Its Applications — Heron's formula finds the area of a triangle from its three sides alone, without needing the height.

## Chapter 12 — End of Chapter Revision

### All Memory Hooks

- Heron's Formula and Its Applications: 'Half the perimeter, then s minus each side, multiply all four, square root.'

### All Common Mistakes

-  Using the full perimeter instead of the semi-perimeter.
-  Forgetting to take the square root at the end.
-  Rounding intermediate values, which distorts the final answer.

### Formula Sheet

| Formula                                                                                        | Meaning / Use                                                           |
|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>Heron's Formula:</b> $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$ ,<br>where $s = (a + b + c)/2$ | Whenever all three sides are known and the height is not.               |
| <b>Area of an Equilateral Triangle:</b> $\text{Area} = (\sqrt{3} / 4) a^2$                     | For any equilateral triangle, as a shortcut instead of Heron's formula. |

### Quick Revision Section

- Heron's Formula and Its Applications: Heron's formula finds the area of a triangle from its three sides alone, without needing the height.

## Chapter 13: Surface Areas and Volumes of Solids

### Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Area of a rectangle, triangle and circle.
- Idea of three-dimensional shapes and their nets.
- Value of  $\pi$  as  $22/7$  or  $3.14$ .
- Ability to handle squares, cubes and square roots.

### 13.1 Cuboid, Cube and Cylinder

 **Estimated time:** 16 mins   **Difficulty:**  Medium   **★★☆**   **Exam Importance:**     

#### Quick Summary

Surface area is the area of the outer covering of a solid and volume is the space it occupies; for prisms and cylinders volume is simply base area times height.

#### Real-Life Example

The paper needed to wrap a gift box is its surface area, while the amount it can hold is its volume — two different questions about the same box.

### Detailed Explanation

#### Definition

Total surface area is the area of all the faces of a solid. Lateral or curved surface area excludes the top and bottom faces. Volume is the space enclosed by the solid.

#### Key Idea

For any prism or cylinder, volume = base area  $\times$  height. Learning this one principle replaces three separate volume formulas.

#### Working Principle

The curved surface of a cylinder unrolls into a rectangle of width  $2\pi r$  and height  $h$ , which is exactly why its curved surface area is  $2\pi rh$ .

#### Applications

- Cost of painting a wall uses surface area; capacity of a tank uses volume.
- $1 \text{ m}^3 = 1000 \text{ litres}$ ;  $1 \text{ cm}^3 = 1 \text{ millilitre}$ .
- A hollow cylinder such as a pipe requires both the inner and outer radii.

#### Points to Remember (Exam Focus)

- ✓ Cuboid:  $TSA = 2(lb + bh + hl)$ , Volume =  $lbh$ , diagonal =  $\sqrt{l^2 + b^2 + h^2}$ .
- ✓ Cube:  $TSA = 6a^2$ , Volume =  $a^3$ , diagonal =  $a\sqrt{3}$ .
- ✓ Cylinder:  $CSA = 2\pi rh$ ,  $TSA = 2\pi r(r + h)$ , Volume =  $\pi r^2 h$ .
- ✓ Lateral surface area of a cuboid =  $2h(l + b)$ , excluding top and bottom.
- ✓ All dimensions must be converted to the same unit before substituting.

### ⚠ Common Student Mistakes

- ⚠ Using total surface area where only the curved or lateral area is needed (e.g. painting only the walls).
- ⚠ Mixing units — length in metres and radius in centimetres.
- ⚠ Using the diameter in place of the radius in a cylinder formula.



### Memory Hook

'Volume = base area  $\times$  height' for every prism and cylinder — one rule, three solids.



### Advanced Insights

Cavalieri's principle explains why a slanted stack of coins has the same volume as a straight one: equal cross-sections at every height give equal volume, regardless of how the solid leans.

## Formula Section



### Cuboid and Cube

|                         |                                                                              |
|-------------------------|------------------------------------------------------------------------------|
| Difficulty              | ● Easy ★☆☆                                                                   |
| Formula                 | Cuboid: $V = lbh$ , $TSA = 2(lb + bh + hl)$ ; Cube: $V = a^3$ , $TSA = 6a^2$ |
| Meaning of Variables    | $l, b, h$ = length, breadth, height; $a$ = edge of the cube.                 |
| When to Use             | For boxes, rooms, tanks, bricks and any rectangular solid.                   |
| Common Mistakes         | Forgetting the factor 2 in the total surface area of a cuboid.               |
| Shortcut / Memory Trick | Diagonal of a cuboid = $\sqrt{l^2 + b^2 + h^2}$ ; of a cube = $a\sqrt{3}$ .  |



### Cylinder

|                      |                                                           |
|----------------------|-----------------------------------------------------------|
| Difficulty           | ● Medium ★★☆☆                                             |
| Formula              | $CSA = 2\pi rh$ , $TSA = 2\pi r(r + h)$ , $V = \pi r^2 h$ |
| Meaning of Variables | $r$ = radius of the base, $h$ = height of the cylinder.   |

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>When to Use</b>             | For pipes, tanks, rollers, cans and drums.                                                  |
| <b>Common Mistakes</b>         | Substituting the diameter for $r$ , which quadruples the volume.                            |
| <b>Shortcut / Memory Trick</b> | A roller covers CSA per revolution; multiply by the number of revolutions for area covered. |

## 13.2 Cone, Sphere and Hemisphere

🕒 **Estimated time:** 16 mins   **Difficulty:** ● Medium   ★★☆☆   **Exam Importance:** ★★★★★

### 🔍 Quick Summary

A cone has one third the volume of a cylinder on the same base and height, and a sphere has volume  $(4/3)\pi r^3$  with surface area  $4\pi r^2$ .

### 🌍 Real-Life Example

An ice cream cone holds exactly one third of what a cylindrical cup of the same width and height would hold — which is why it looks bigger than it is.

### 📖 Detailed Explanation

#### Definition

The slant height  $l$  of a cone is the distance from the apex to any point on the edge of the base, and satisfies  $l^2 = r^2 + h^2$ . A hemisphere is half of a sphere.

#### Key Idea

The cone's curved surface uses the slant height  $l$ , while its volume uses the vertical height  $h$ . Substituting one for the other is the most frequent error in this chapter.

#### Working Principle

For combined solids, add the curved surfaces of the parts that remain exposed and ignore the joined faces. Volumes always add directly.

#### Applications

- A cone, hemisphere and cylinder of the same radius and height have volumes in the ratio 1 : 2 : 3.
- A sphere's surface area equals the curved surface of the cylinder that just encloses it.
- Melting and recasting problems keep volume constant while shape changes.

#### ✓ Points to Remember (Exam Focus)

- ✓ Cone:  $l^2 = r^2 + h^2$ ,  $CSA = \pi r l$ ,  $TSA = \pi r(l + r)$ ,  $V = (1/3)\pi r^2 h$ .
- ✓ Sphere: Surface area =  $4\pi r^2$ ,  $V = (4/3)\pi r^3$ .
- ✓ Hemisphere:  $CSA = 2\pi r^2$ ,  $TSA = 3\pi r^2$ ,  $V = (2/3)\pi r^3$ .
- ✓ Cone : hemisphere : cylinder volumes on the same base and height = 1 : 2 : 3.

✓ In recasting problems, volume is conserved even though surface area changes.

### ⚠ Common Student Mistakes

- ⚠ Using slant height  $l$  in the volume formula instead of vertical height  $h$ .
- ⚠ Taking the total surface area of a hemisphere as  $2\pi r^2$  and forgetting the flat circular face.
- ⚠ Adding the surface areas of joined faces in a combined solid.



### Memory Hook

'Cone is a third of a cylinder, sphere is two thirds of it' — and 'l for curved, h for volume'.



### Advanced Insights

Archimedes proved that a sphere has two thirds the volume and two thirds the total surface area of its circumscribing cylinder, and asked for that figure to be carved on his tomb — he considered it his finest result.

## Formula Section



### Cone

|                         |                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------|
| Difficulty              | ● Medium ★★☆☆                                                                         |
| Formula                 | $l = \sqrt{r^2 + h^2}$ , $CSA = \pi rl$ , $TSA = \pi r(l + r)$ , $V = (1/3)\pi r^2 h$ |
| Meaning of Variables    | $r$ = base radius, $h$ = vertical height, $l$ = slant height.                         |
| When to Use             | For conical tents, funnels, heaps of grain and ice-cream cones.                       |
| Common Mistakes         | Using $l$ in the volume formula, or $h$ in the curved surface area formula.           |
| Shortcut / Memory Trick | Canvas needed for a conical tent is the curved surface area only.                     |



### Sphere and Hemisphere

|                      |                                                                                                                     |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Difficulty           | ● Medium ★★☆☆                                                                                                       |
| Formula              | Sphere: $SA = 4\pi r^2$ , $V = (4/3)\pi r^3$ ; Hemisphere: $CSA = 2\pi r^2$ , $TSA = 3\pi r^2$ , $V = (2/3)\pi r^3$ |
| Meaning of Variables | $r$ = radius.                                                                                                       |
| When to Use          | For balls, domes, bowls and spherical tanks.                                                                        |

|                                |                                                                                 |
|--------------------------------|---------------------------------------------------------------------------------|
| <b>Common Mistakes</b>         | Omitting the flat face when the total surface area of a hemisphere is required. |
| <b>Shortcut / Memory Trick</b> | Hemisphere TSA = $2\pi r^2$ (curved) + $\pi r^2$ (flat) = $3\pi r^2$ .          |

## II Quick Recap — Topics Covered So Far

- Cuboid, Cube and Cylinder — Surface area is the area of the outer covering of a solid and volume is the space it occupies; for prisms and cylinders volume is simply base area times height.
- Cone, Sphere and Hemisphere — A cone has one third the volume of a cylinder on the same base and height, and a sphere has volume  $(4/3)\pi r^3$  with surface area  $4\pi r^2$ .

## Chapter 13 — End of Chapter Revision

### All Memory Hooks

- Cuboid, Cube and Cylinder: 'Volume = base area  $\times$  height' for every prism and cylinder — one rule, three solids.
- Cone, Sphere and Hemisphere: 'Cone is a third of a cylinder, sphere is two thirds of it' — and 'l for curved, h for volume'.

### All Common Mistakes

-  Using total surface area where only the curved or lateral area is needed (e.g. painting only the walls).
-  Mixing units — length in metres and radius in centimetres.
-  Using the diameter in place of the radius in a cylinder formula.
-  Using slant height  $l$  in the volume formula instead of vertical height  $h$ .
-  Taking the total surface area of a hemisphere as  $2\pi r^2$  and forgetting the flat circular face.
-  Adding the surface areas of joined faces in a combined solid.

### Formula Sheet

| Formula                                                                                                                                           | Meaning / Use                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>Cuboid and Cube:</b> Cuboid: $V = lbh$ , $TSA = 2(lb+bh+hl)$ ; Cube: $V = a^3$ , $TSA = 6a^2$                                                  | For boxes, rooms, tanks, bricks and any rectangular solid.      |
| <b>Cylinder:</b> $CSA = 2\pi rh$ , $TSA = 2\pi r(r + h)$ , $V = \pi r^2 h$                                                                        | For pipes, tanks, rollers, cans and drums.                      |
| <b>Cone:</b> $l = \sqrt{r^2 + h^2}$ , $CSA = \pi rl$ , $TSA = \pi r(l + r)$ , $V = (1/3)\pi r^2 h$                                                | For conical tents, funnels, heaps of grain and ice-cream cones. |
| <b>Sphere and Hemisphere:</b> Sphere: $SA = 4\pi r^2$ , $V = (4/3)\pi r^3$ ; Hemisphere: $CSA = 2\pi r^2$ , $TSA = 3\pi r^2$ , $V = (2/3)\pi r^3$ | For balls, domes, bowls and spherical tanks.                    |

### Quick Revision Section

- Cuboid, Cube and Cylinder: Surface area is the area of the outer covering of a solid and volume is the space it occupies; for prisms and cylinders volume is simply base area times height.
- Cone, Sphere and Hemisphere: A cone has one third the volume of a cylinder on the same base and height, and a sphere has volume  $(4/3)\pi r^3$  with surface area  $4\pi r^2$ .



## Chapter 14: Statistics

### Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Ability to calculate an average of a set of numbers.
- Familiarity with tables and bar graphs.
- Basic understanding of fractions and percentages.
- Skill in plotting points on a graph.

### 14.1 Collection and Presentation of Data

 **Estimated time:** 14 mins   **Difficulty:**  Easy ★☆☆ **Exam Importance:** ★★★★★

#### Quick Summary

Raw data is organised into frequency distributions and displayed as bar graphs, histograms or frequency polygons to make patterns visible.

#### Real-Life Example

A histogram of a class's marks shows at a glance whether the paper was easy or hard — something a list of eighty numbers never could.

### Detailed Explanation

#### Definition

Primary data is collected by the investigator directly; secondary data is obtained from an existing source. Class size is the difference between the upper and lower limits of a class, and the class mark is their average.

#### Key Idea

A histogram is not a bar graph. Histogram bars touch because the variable is continuous, whereas bar graph bars are separated because the categories are discrete.

#### Working Principle

For a grouped frequency distribution with unequal class sizes, the height of each histogram bar must be adjusted so that area, not height, represents frequency.

#### Applications

- Discontinuous classes such as 10–19, 20–29 are made continuous by subtracting 0.5 from lower limits and adding 0.5 to upper limits.
- A frequency polygon is drawn by joining the mid-points of the tops of histogram bars.
- Class mark =  $(\text{upper limit} + \text{lower limit}) / 2$ .

**✓ Points to Remember (Exam Focus)**

- ✓ Primary data is first-hand; secondary data comes from an existing source.
- ✓ Range = maximum value – minimum value.
- ✓ Class mark = (upper limit + lower limit) / 2.
- ✓ In a histogram the bars touch; in a bar graph they do not.
- ✓ With unequal class sizes, the AREA of each bar must be proportional to the frequency.

**⚠ Common Student Mistakes**

- ⚠ Drawing a histogram with gaps between the bars.
- ⚠ Using raw frequencies as heights when class sizes are unequal.
- ⚠ Forgetting to convert discontinuous classes into continuous ones before drawing.

**Memory Hook**

'Histogram bars touch, bar-graph bars do not' — continuous versus categorical.

**Advanced Insights**

Choosing the class width is a genuine trade-off: too wide hides structure, too narrow shows noise. Statisticians use rules such as Sturges' formula, but the honest answer is that several widths should be tried.

## 14.2 Mean, Median and Mode

**Estimated time:** 15 mins   **Difficulty:** Medium   ★★☆☆   **Exam Importance:** ★★★★★

**Quick Summary**

The mean is the arithmetic average, the median is the middle value when data is arranged in order, and the mode is the most frequent value.

**Real-Life Example**

Average income in a town is misleading if one billionaire lives there — the median gives a far truer picture of a typical resident.

**Detailed Explanation****Definition**

Mean = sum of observations ÷ number of observations. Median is the middle observation of ordered data. Mode is the observation occurring most often.

**Key Idea**

The mean is pulled by extreme values while the median is not. Choosing between them is a judgement about whether outliers are meaningful or distorting.

**Working Principle**

For  $n$  observations arranged in order, the median is the  $(n+1)/2$  th value when  $n$  is odd, and the average of the  $n/2$  th and  $(n/2 + 1)$ th values when  $n$  is even.

**Applications**

- The median is preferred for income, house prices and any skewed data.
- The mode is the only average usable for non-numerical data such as favourite colour.
- Empirical relation:  $\text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean}$ .

**✓ Points to Remember (Exam Focus)**

- ✓ Mean =  $\Sigma x / n$ ; for grouped data, mean =  $\Sigma fx / \Sigma f$ .
- ✓ Median: order the data first — this step is essential and often forgotten.
- ✓ Mode is the most frequent value; a data set may have more than one mode or none.
- ✓ The mean is affected by extreme values; the median is not.
- ✓ Empirical relation:  $\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$ .

**⚠ Common Student Mistakes**

- ⚠ Finding the median without first arranging the data in order.
- ⚠ Using  $n/2$  as the median position for odd  $n$  instead of  $(n+1)/2$ .
- ⚠ Reporting the frequency instead of the observation as the mode.

**🧠 Memory Hook**

'Mean adds, Median orders, Mode counts' — three different questions about the same data.

**🚀 Advanced Insights**

Mean, median and mode coincide in a perfectly symmetric distribution. The size and direction of the gap between them is a standard measure of skewness — which is what the empirical relation is really capturing.

**Formula Section****📊 Mean of Grouped Data**

|                   |                                              |
|-------------------|----------------------------------------------|
| <b>Difficulty</b> | ● Medium ★★☆☆                                |
| <b>Formula</b>    | $\text{Mean} = \Sigma(f_i x_i) / \Sigma f_i$ |

|                                |                                                                                |
|--------------------------------|--------------------------------------------------------------------------------|
| <b>Meaning of Variables</b>    | $f_i$ = frequency of the $i$ th class, $x_i$ = class mark of the $i$ th class. |
| <b>When to Use</b>             | For any grouped frequency distribution.                                        |
| <b>Common Mistakes</b>         | Using the class limits instead of the class marks as $x_i$ .                   |
| <b>Shortcut / Memory Trick</b> | The assumed-mean method reduces arithmetic when the class marks are large.     |

### Median

|                                |                                                                                     |
|--------------------------------|-------------------------------------------------------------------------------------|
| <b>Difficulty</b>              | ● Easy ★☆☆                                                                          |
| <b>Formula</b>                 | $n$ odd: $((n+1)/2)$ th term; $n$ even: mean of $(n/2)$ th and $(n/2 + 1)$ th terms |
| <b>Meaning of Variables</b>    | $n$ = number of observations, arranged in ascending order.                          |
| <b>When to Use</b>             | When the data contains extreme values that would distort the mean.                  |
| <b>Common Mistakes</b>         | Failing to sort the data before locating the middle term.                           |
| <b>Shortcut / Memory Trick</b> | Sort first, always — the formula is meaningless on unsorted data.                   |

## II Quick Recap — Topics Covered So Far

- Collection and Presentation of Data — Raw data is organised into frequency distributions and displayed as bar graphs, histograms or frequency polygons to make patterns visible.
- Mean, Median and Mode — The mean is the arithmetic average, the median is the middle value when data is arranged in order, and the mode is the most frequent value.

## Chapter 14 — End of Chapter Revision

### All Memory Hooks

- Collection and Presentation of Data: 'Histogram bars touch, bar-graph bars do not' — continuous versus categorical.
- Mean, Median and Mode: 'Mean adds, Median orders, Mode counts' — three different questions about the same data.

### All Common Mistakes

-  Drawing a histogram with gaps between the bars.
-  Using raw frequencies as heights when class sizes are unequal.
-  Forgetting to convert discontinuous classes into continuous ones before drawing.
-  Finding the median without first arranging the data in order.
-  Using  $n/2$  as the median position for odd  $n$  instead of  $(n+1)/2$ .
-  Reporting the frequency instead of the observation as the mode.

### Formula Sheet

| Formula                                                                                            | Meaning / Use                                                      |
|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Mean of Grouped Data:</b> $\text{Mean} = \frac{\sum(f_i x_i)}{\sum f_i}$                        | For any grouped frequency distribution.                            |
| <b>Median:</b> $n$ odd: $((n+1)/2)$ th term; $n$ even: mean of $(n/2)$ th and $(n/2 + 1)$ th terms | When the data contains extreme values that would distort the mean. |

### Quick Revision Section

- Collection and Presentation of Data: Raw data is organised into frequency distributions and displayed as bar graphs, histograms or frequency polygons to make patterns visible.
- Mean, Median and Mode: The mean is the arithmetic average, the median is the middle value when data is arranged in order, and the mode is the most frequent value.

## Chapter 15: Probability

### Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Ability to work with fractions and percentages.
- Idea of counting favourable cases out of total cases.
- Familiarity with dice, coins and playing cards.
- Understanding of a ratio.

### 15.1 Empirical Probability

 **Estimated time:** 14 mins   **Difficulty:**  Easy   ★☆☆   **Exam Importance:** ★★★★★

#### Quick Summary

Empirical or experimental probability is the ratio of the number of times an event occurs to the total number of trials, and it approaches the theoretical value as trials increase.

#### Real-Life Example

A weather forecast of 70% rain comes from records of past days with similar conditions — empirical probability at work.

#### Detailed Explanation

##### Definition

The empirical probability of an event  $E$  is  $P(E) = (\text{number of trials in which } E \text{ happened}) / (\text{total number of trials})$ . A trial is one performance of an experiment.

##### Key Idea

Empirical probability is based on actual observation and can change with each new set of trials, whereas theoretical probability is computed from the structure of the experiment and does not.

##### Working Principle

The law of large numbers guarantees that as the number of trials grows, empirical probability settles closer and closer to the theoretical value.

##### Applications

- Insurance premiums are computed from empirical probabilities of claims.
- Quality control estimates defect rates by sampling.
- Medical trials estimate the probability of a treatment succeeding.

#### ✓ Points to Remember (Exam Focus)

- ✓  $P(E)$  = favourable trials / total trials.
- ✓  $0 \leq P(E) \leq 1$  always.
- ✓  $P(E) = 0$  means an impossible event;  $P(E) = 1$  means a certain event.
- ✓ The sum of the probabilities of all elementary outcomes is 1.
- ✓  $P(\text{not } E) = 1 - P(E)$ .

### ⚠ Common Student Mistakes

- ⚠ Reporting a probability greater than 1 or negative.
- ⚠ Dividing by the number of favourable cases instead of the total.
- ⚠ Assuming a small number of trials gives a reliable estimate.

### 🧠 Memory Hook

**'Favourable over Total — and the answer can never leave the range 0 to 1.'**

### 🚀 Advanced Insights

The gambler's fallacy — believing a coin is 'due' to land heads after several tails — comes from confusing the law of large numbers with a short-run correction. Independent trials have no memory whatsoever.

## II Quick Recap — Topics Covered So Far

- Empirical Probability — Empirical or experimental probability is the ratio of the number of times an event occurs to the total number of trials, and it approaches the theoretical value as trials increase.

## Chapter 15 — End of Chapter Revision

### All Memory Hooks

- Empirical Probability: 'Favourable over Total — and the answer can never leave the range 0 to 1.'

### All Common Mistakes

-  Reporting a probability greater than 1 or negative.
-  Dividing by the number of favourable cases instead of the total.
-  Assuming a small number of trials gives a reliable estimate.

### Quick Revision Section

- Empirical Probability: Empirical or experimental probability is the ratio of the number of times an event occurs to the total number of trials, and it approaches the theoretical value as trials increase.

## Chapter 16: Linear Inequalities

### Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Ability to solve linear equations in one variable.
- Familiarity with the number line.
- Understanding of the symbols  $<$ ,  $>$ ,  $\leq$  and  $\geq$ .
- Knowledge of the Cartesian plane.

### 16.1 Solving Linear Inequalities in One Variable

 **Estimated time:** 15 mins   **Difficulty:**  Medium   **★★☆**   **Exam Importance:**     

#### Quick Summary

Linear inequalities are solved like equations, with one crucial exception: multiplying or dividing by a negative number reverses the inequality sign.

#### Real-Life Example

A lift rated 'not more than 600 kg' defines an inequality, and every safety limit in engineering is written the same way.

#### Detailed Explanation

##### Definition

A strict inequality uses  $<$  or  $>$ ; a slack inequality uses  $\leq$  or  $\geq$ . The solution set is the set of all values of the variable satisfying the inequality.

##### Key Idea

Adding or subtracting the same number leaves an inequality unchanged, and so does multiplying or dividing by a positive number. Only a negative multiplier flips the sign — that is the entire difference from equation solving.

##### Working Principle

The solution of a linear inequality is generally an interval, not a single value, and is represented on the number line with a hollow circle for a strict inequality and a filled circle for a slack one.

##### Applications

- $-3x > 12$  gives  $x < -4$  after dividing by  $-3$  and reversing the sign.
- Solution sets are written in interval notation, e.g.  $(-\infty, -4)$ .
- Restricting the solution to integers or natural numbers changes the answer set.

#### Points to Remember (Exam Focus)

- ✓ Multiplying or dividing both sides by a **NEGATIVE** number reverses the inequality.
- ✓ Adding or subtracting any number does not change the direction.
- ✓ Strict inequality ( $<$ ,  $>$ ) is shown by a hollow circle on the number line.
- ✓ Slack inequality ( $\leq$ ,  $\geq$ ) is shown by a filled circle.
- ✓ The solution is usually an interval containing infinitely many values.

### ⚠ Common Student Mistakes

- ⚠ Forgetting to reverse the sign after dividing by a negative number.
- ⚠ Giving a single value as the answer instead of an interval.
- ⚠ Using a filled circle for a strict inequality.



### Memory Hook

**'Negative flips it' — divide or multiply by a negative and the inequality turns around.**



### Advanced Insights

Systems of linear inequalities define feasible regions in the plane, which is the foundation of linear programming. Optimising a linear objective over such a region always gives an answer at a corner of the region.

## II Quick Recap — Topics Covered So Far

- Solving Linear Inequalities in One Variable — Linear inequalities are solved like equations, with one crucial exception: multiplying or dividing by a negative number reverses the inequality sign.

## Chapter 16 — End of Chapter Revision

### All Memory Hooks

- Solving Linear Inequalities in One Variable: 'Negative flips it' — divide or multiply by a negative and the inequality turns around.

### All Common Mistakes

-  Forgetting to reverse the sign after dividing by a negative number.
-  Giving a single value as the answer instead of an interval.
-  Using a filled circle for a strict inequality.

### Quick Revision Section

- Solving Linear Inequalities in One Variable: Linear inequalities are solved like equations, with one crucial exception: multiplying or dividing by a negative number reverses the inequality sign.

## Chapter 17: Permutations and Combinations

### Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Ability to multiply large numbers systematically.
- Understanding of the factorial notation.
- Basic counting of possibilities.
- Familiarity with the idea of an ordered pair.

### 17.1 Fundamental Principle of Counting and Permutations

 **Estimated time:** 16 mins   **Difficulty:**  Medium   **★★☆**   **Exam Importance:**     

#### Quick Summary

If one task can be done in  $m$  ways and another in  $n$  ways, together they can be done in  $m \times n$  ways; a permutation is an arrangement in which order matters.

#### Real-Life Example

A four-digit PIN has 10,000 possibilities because each of the four positions independently has ten choices — the multiplication principle securing your bank account.

#### Detailed Explanation

##### Definition

The fundamental principle of counting states that if an event can occur in  $m$  ways and a second independent event in  $n$  ways, then both can occur in  $m \times n$  ways. A permutation is an arrangement of objects in a definite order.

##### Key Idea

Permutations count arrangements, so ABC and BCA are different. This is the one question to ask before choosing a formula: does the order matter?

##### Working Principle

The number of permutations of  $n$  distinct objects taken  $r$  at a time is  ${}^n P_r = n!/(n-r)!$ . When all  $n$  objects are used, this becomes  $n!$ .

##### Applications

- Number of ways to seat 5 people in 5 chairs =  $5! = 120$ .
- Number of 3-letter words from 5 distinct letters =  ${}^5 P_3 = 60$ .
- Circular permutations of  $n$  objects =  $(n-1)!$ .

### ✓ Points to Remember (Exam Focus)

- ✓ Multiplication principle for 'and'; addition principle for 'or'.
- ✓  $n! = n \times (n-1) \times \dots \times 2 \times 1$ , and  $0! = 1$  by definition.
- ✓  ${}^n P_r = n! / (n-r)!$ .
- ✓ Permutations of  $n$  objects where  $p$  are alike  $= n!/p!$ .
- ✓ Circular permutations of  $n$  distinct objects  $= (n-1)!$ .

### ⚠ Common Student Mistakes

- ⚠ Taking  $0!$  as  $0$ ; it is defined to be  $1$ .
- ⚠ Adding when the tasks must both be done — 'and' means multiply.
- ⚠ Using  $n!$  for circular arrangements instead of  $(n-1)!$ .



### Memory Hook

'AND means multiply, OR means add' — and 'Permutation = Position matters'.



### Advanced Insights

Circular arrangements lose one factor of  $n$  because rotating the whole circle produces the same arrangement. Whenever a symmetry maps arrangements onto each other, the count must be divided by the size of that symmetry group.

## Formula Section



### Permutations

|                         |                                                                                  |
|-------------------------|----------------------------------------------------------------------------------|
| Difficulty              | ● Medium ★★☆☆                                                                    |
| Formula                 | ${}^n P_r = n! / (n-r)!$                                                         |
| Meaning of Variables    | $n$ = total number of distinct objects, $r$ = number of objects taken at a time. |
| When to Use             | When order matters — arrangements, rankings, seatings, words from letters.       |
| Common Mistakes         | Using it where order does not matter, which requires combinations instead.       |
| Shortcut / Memory Trick | ${}^n P_n = n!$ and ${}^n P_1 = n$ .                                             |

## 17.2 Combinations

🕒 Estimated time: 14 mins    Difficulty: ● Medium ★★☆☆    Exam Importance: ★★★★★

### Quick Summary

A combination is a selection in which order does not matter, and  ${}^nC_r = n!/[r!(n-r)!]$ .

### Real-Life Example

A cricket selection committee choosing 11 from 15 players is making a combination — who is picked matters, the order of picking does not.

## Detailed Explanation

### Definition

A combination is a selection of objects without regard to order. The number of combinations of  $n$  distinct objects taken  $r$  at a time is  ${}^nC_r = n!/[r!(n-r)!]$ .

### Key Idea

Every combination of  $r$  objects can be arranged in  $r!$  ways, so  ${}^nP_r = {}^nC_r \times r!$ . Dividing permutations by  $r!$  removes the ordering, and that is the whole relationship between the two.

### Working Principle

The symmetry  ${}^nC_r = {}^nC_{n-r}$  holds because choosing  $r$  objects to include is the same as choosing  $n - r$  to leave out. It often halves the arithmetic.

### Applications

- Selecting a committee, a team or a hand of cards.
- ${}^nC_0 = {}^nC_n = 1$  and  ${}^nC_1 = n$ .
- The number of subsets of a set with  $n$  elements is  $2^n$ .

### ✓ Points to Remember (Exam Focus)

- ✓  ${}^nC_r = n!/[r!(n-r)!]$ .
- ✓  ${}^nP_r = {}^nC_r \times r!$ .
- ✓  ${}^nC_r = {}^nC_{n-r}$  — choosing what to take is the same as choosing what to leave.
- ✓  ${}^nC_0 = {}^nC_n = 1$ ;  ${}^nC_1 = n$ .
- ✓ Sum of all  ${}^nC_r$  from  $r = 0$  to  $n$  equals  $2^n$ .

### Common Student Mistakes

-  Using combinations for arrangements, where order matters.
-  Forgetting the  $r!$  in the denominator, which gives permutations instead.
-  Computing full factorials of large  $n$  instead of cancelling first.

### Memory Hook

'Combination = Committee (order irrelevant); Permutation = Podium (order decides everything).'

### Advanced Insights

The numbers  ${}^nC_r$  are exactly the entries of Pascal's triangle and the coefficients in the binomial expansion of  $(a + b)^n$ . Counting, algebra and the triangle are three views of one object.

## Formula Section

### Combinations

|                         |                                                                              |
|-------------------------|------------------------------------------------------------------------------|
| Difficulty              | ● Medium ★★☆☆                                                                |
| Formula                 | ${}^nC_r = n! / [r! (n - r)!]$                                               |
| Meaning of Variables    | $n$ = total objects, $r$ = objects selected.                                 |
| When to Use             | When order does not matter — teams, committees, subsets, card hands.         |
| Common Mistakes         | Omitting $r!$ in the denominator, which turns the answer into a permutation. |
| Shortcut / Memory Trick | Use ${}^nC_r = {}^nC_{n-r}$ to pick whichever of $r$ or $n-r$ is smaller.    |

## II Quick Recap — Topics Covered So Far

- Fundamental Principle of Counting and Permutations — If one task can be done in  $m$  ways and another in  $n$  ways, together they can be done in  $m \times n$  ways; a permutation is an arrangement in which order matters.
- Combinations — A combination is a selection in which order does not matter, and  ${}^nC_r = n! / [r!(n-r)!]$ .

## Chapter 17 — End of Chapter Revision

### All Memory Hooks

- Fundamental Principle of Counting and Permutations: 'AND means multiply, OR means add' — and 'Permutation = Position matters'.
- Combinations: 'Combination = Committee (order irrelevant); Permutation = Podium (order decides everything).'

### All Common Mistakes

-  Taking  $0!$  as  $0$ ; it is defined to be  $1$ .
-  Adding when the tasks must both be done — 'and' means multiply.
-  Using  $n!$  for circular arrangements instead of  $(n - 1)!$ .
-  Using combinations for arrangements, where order matters.
-  Forgetting the  $r!$  in the denominator, which gives permutations instead.
-  Computing full factorials of large  $n$  instead of cancelling first.

### Formula Sheet

| Formula                                              | Meaning / Use                                                              |
|------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Permutations:</b> ${}^n P_r = n! / (n - r)!$      | When order matters — arrangements, rankings, seatings, words from letters. |
| <b>Combinations:</b> ${}^n C_r = n! / [r! (n - r)!]$ | When order does not matter — teams, committees, subsets, card hands.       |

### Quick Revision Section

- Fundamental Principle of Counting and Permutations: If one task can be done in  $m$  ways and another in  $n$  ways, together they can be done in  $m \times n$  ways; a permutation is an arrangement in which order matters.
- Combinations: A combination is a selection in which order does not matter, and  ${}^n C_r = n!/[r!(n-r)!]$ .

## Chapter 18: Limits

### Prerequisites

Before starting this chapter, make sure you are comfortable with the following ideas:

- Ability to substitute a value into an algebraic expression.
- Skill in factorising polynomials.
- Understanding of rational expressions.
- Familiarity with the idea of a function.

### 18.1 Concept of a Limit and Its Evaluation

 **Estimated time:** 16 mins   **Difficulty:**  Hard   **★★★**   **Exam Importance:**    

#### Quick Summary

The limit of a function as  $x$  approaches a value is the value the function gets arbitrarily close to, which need not equal the function's value at that point.

#### Real-Life Example

A car's speedometer reads instantaneous speed — the limit of average speed as the time interval shrinks towards zero.

#### Detailed Explanation

##### Definition

$\lim_{x \rightarrow a} f(x) = L$  means that  $f(x)$  can be made as close to  $L$  as desired by taking  $x$  sufficiently close to  $a$ , but not equal to  $a$ .

##### Key Idea

A limit describes the approach, not the arrival. The function need not even be defined at  $x = a$  for the limit there to exist — which is exactly why limits are more powerful than substitution.

##### Working Principle

If direct substitution gives  $0/0$ , the expression is indeterminate and must be simplified by factorising, rationalising or using a standard limit before substituting again.

##### Applications

- $\lim_{x \rightarrow 2} (x^2 - 4)/(x - 2) = 4$ , found by factorising to  $(x + 2)$ .
- The derivative is defined as a limit of a difference quotient.
- Left-hand and right-hand limits must be equal for the limit to exist.

#### Points to Remember (Exam Focus)

- ✓ The limit exists only if the left-hand and right-hand limits are equal.
- ✓ The limit at a point does not depend on the function's value at that point.
- ✓  $0/0$  is an indeterminate form and needs algebraic simplification.
- ✓  $\lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim f(x) \pm \lim g(x)$ , and similarly for products and quotients.
- ✓ Standard limit:  $\lim_{x \rightarrow a} (x^n - a^n)/(x - a) = n \cdot a^{n-1}$ .

### ⚠ Common Student Mistakes

- ⚠ Writing  $0/0 = 1$  or  $0/0 = 0$ ; it is indeterminate, not a number.
- ⚠ Assuming the limit always equals the function's value at that point.
- ⚠ Applying the quotient rule for limits when the denominator's limit is zero.



### Memory Hook

'A limit is about the journey, not the destination' — where the function is heading, not where it lands.



### Advanced Insights

The formal  $\epsilon$ - $\delta$  definition makes 'arbitrarily close' precise and is what put calculus on a rigorous footing in the nineteenth century. Every theorem of calculus ultimately rests on it.

## Formula Section



### Standard Algebraic Limit

|                         |                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------|
| Difficulty              | ● Medium ★★☆☆                                                                       |
| Formula                 | $\lim_{x \rightarrow a} (x^n - a^n)/(x - a) = n \cdot a^{n-1}$                      |
| Meaning of Variables    | $n$ = any rational number, $a$ = the point of approach.                             |
| When to Use             | Whenever a limit produces the indeterminate form $0/0$ with a difference of powers. |
| Common Mistakes         | Applying it when the denominator is not exactly $(x - a)$ .                         |
| Shortcut / Memory Trick | It is the derivative of $x^n$ at $x = a$ , written as a limit.                      |

## II Quick Recap — Topics Covered So Far

- Concept of a Limit and Its Evaluation — The limit of a function as  $x$  approaches a value is the value the function gets arbitrarily close to, which need not equal the function's value at that point.



## Chapter 18 — End of Chapter Revision

### All Memory Hooks

- Concept of a Limit and Its Evaluation: 'A limit is about the journey, not the destination' — where the function is heading, not where it lands.

### All Common Mistakes

-  Writing  $0/0 = 1$  or  $0/0 = 0$ ; it is indeterminate, not a number.
-  Assuming the limit always equals the function's value at that point.
-  Applying the quotient rule for limits when the denominator's limit is zero.

### Formula Sheet

| Formula                                                                                         | Meaning / Use                                                                       |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Standard Algebraic Limit:</b> $\lim_{x \rightarrow a} (x^n - a^n)/(x - a) = n \cdot a^{n-1}$ | Whenever a limit produces the indeterminate form $0/0$ with a difference of powers. |

### Quick Revision Section

- Concept of a Limit and Its Evaluation: The limit of a function as  $x$  approaches a value is the value the function gets arbitrarily close to, which need not equal the function's value at that point.